

Lecture 17

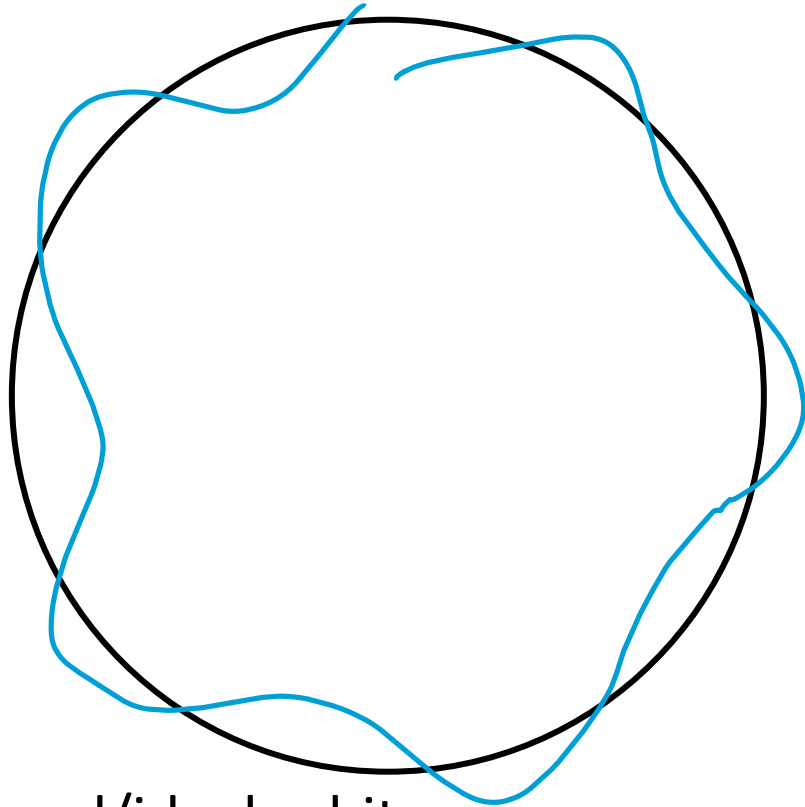
Thursday 7/23/26

Sami Lewis

US Particle Accelerator School
Accelerator Physics – Summer 2026

Reminders about linear transverse motion

Betatron oscillation



Closed/ideal orbit

Poincaré Map: plotted at fixed s location, evolution of the coordinate of particle turn by turn

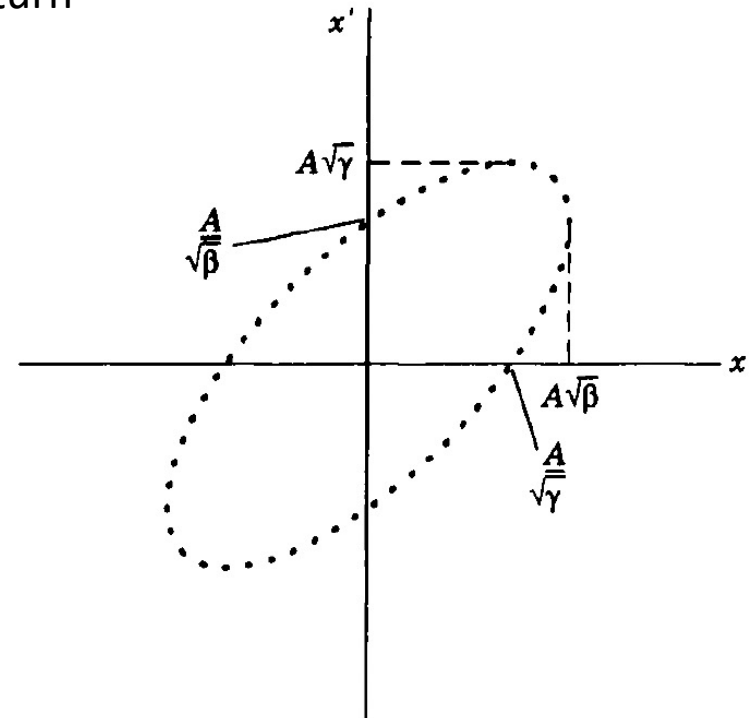
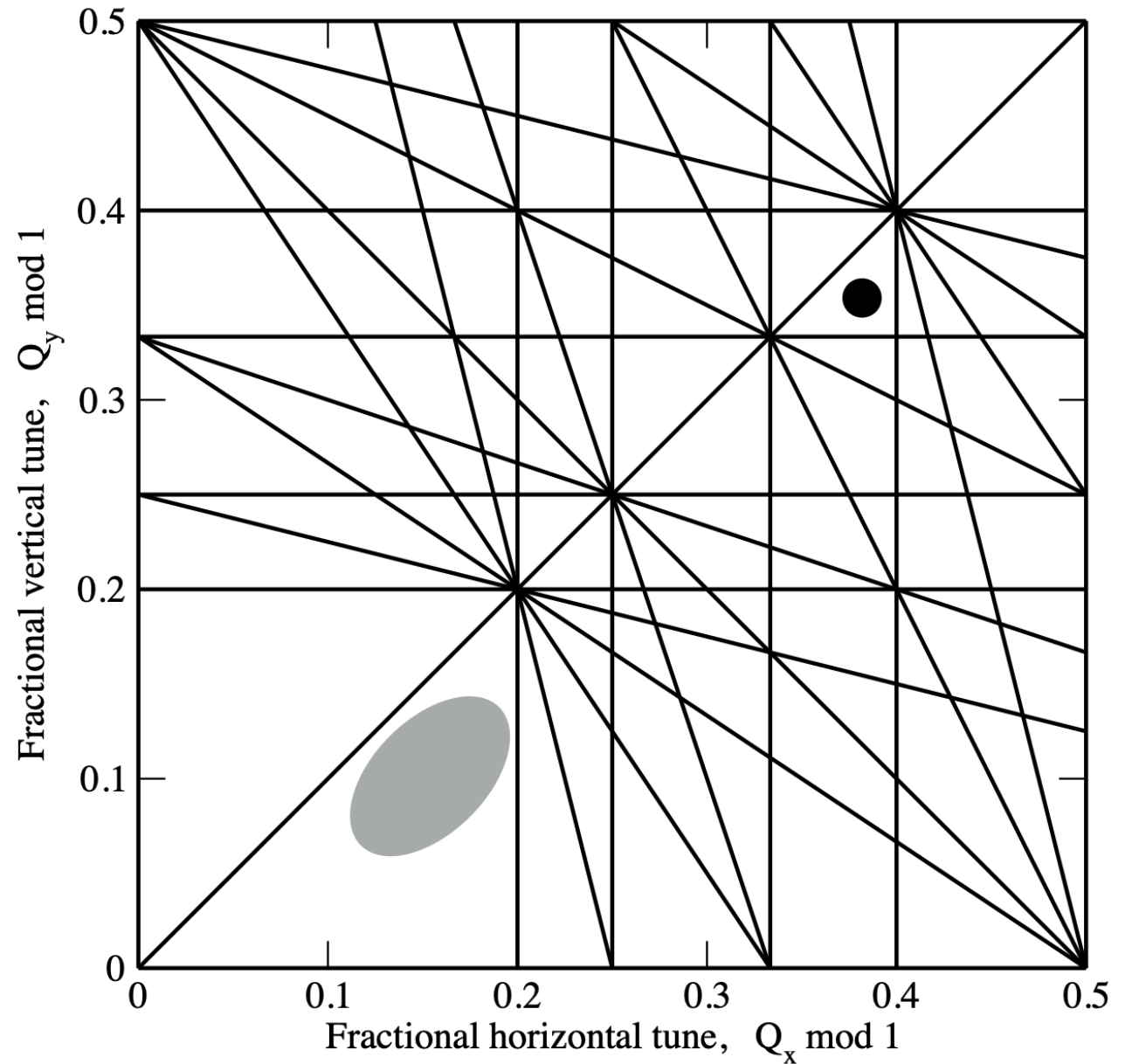


Figure 3.12. Phase space mapping from turn to turn in a circular accelerator.

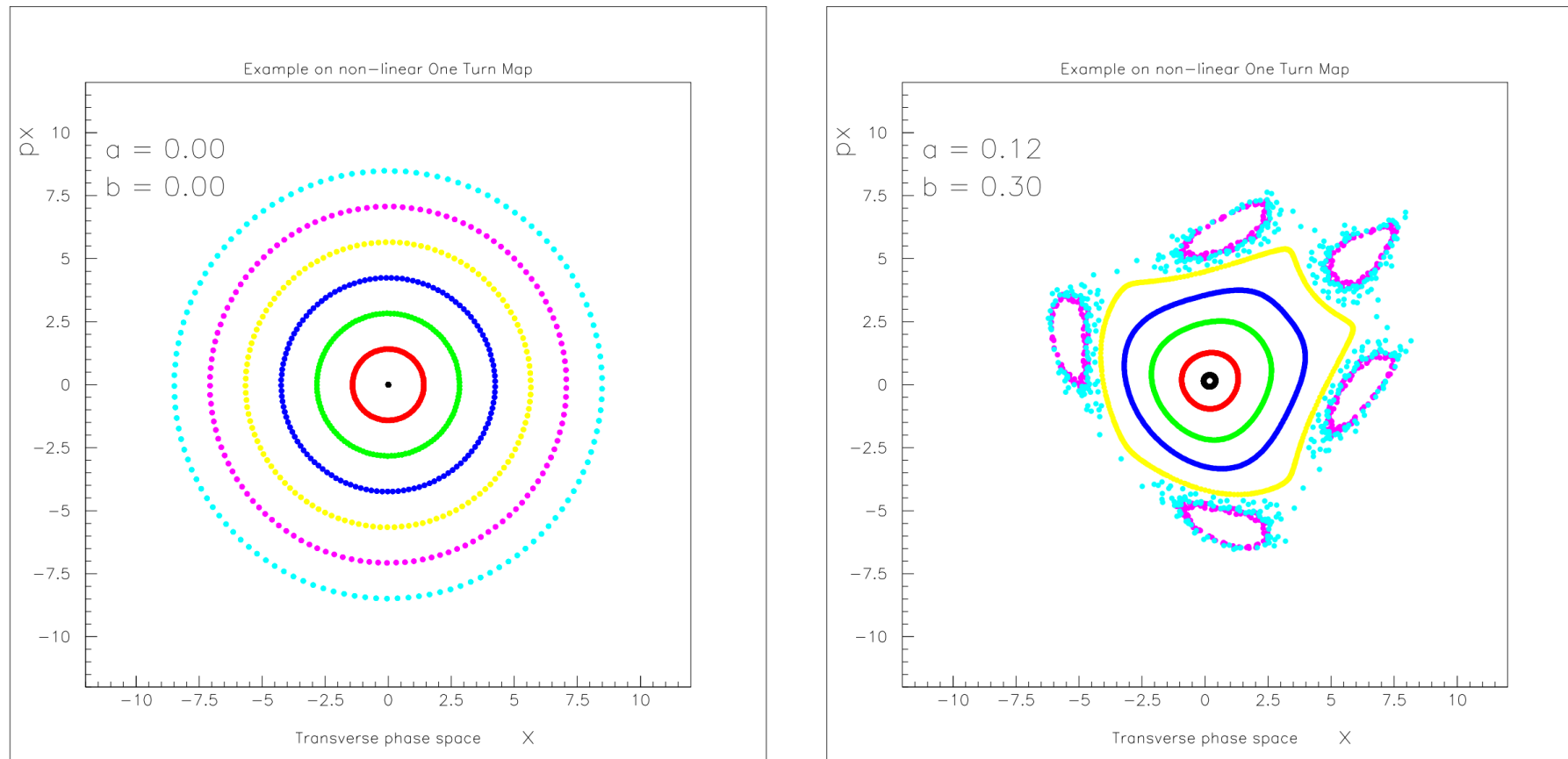
Tune plane

- Resonances resulting in instability/beam loss plotted for horizontal vs vertical tune
- Shaded regions are possible operating ranges
- Gray has much more tune spread from uncorrected chromaticity, compared to black spot



From S. Peggs and T. Satogata, *Introduction to Accelerator Dynamics*

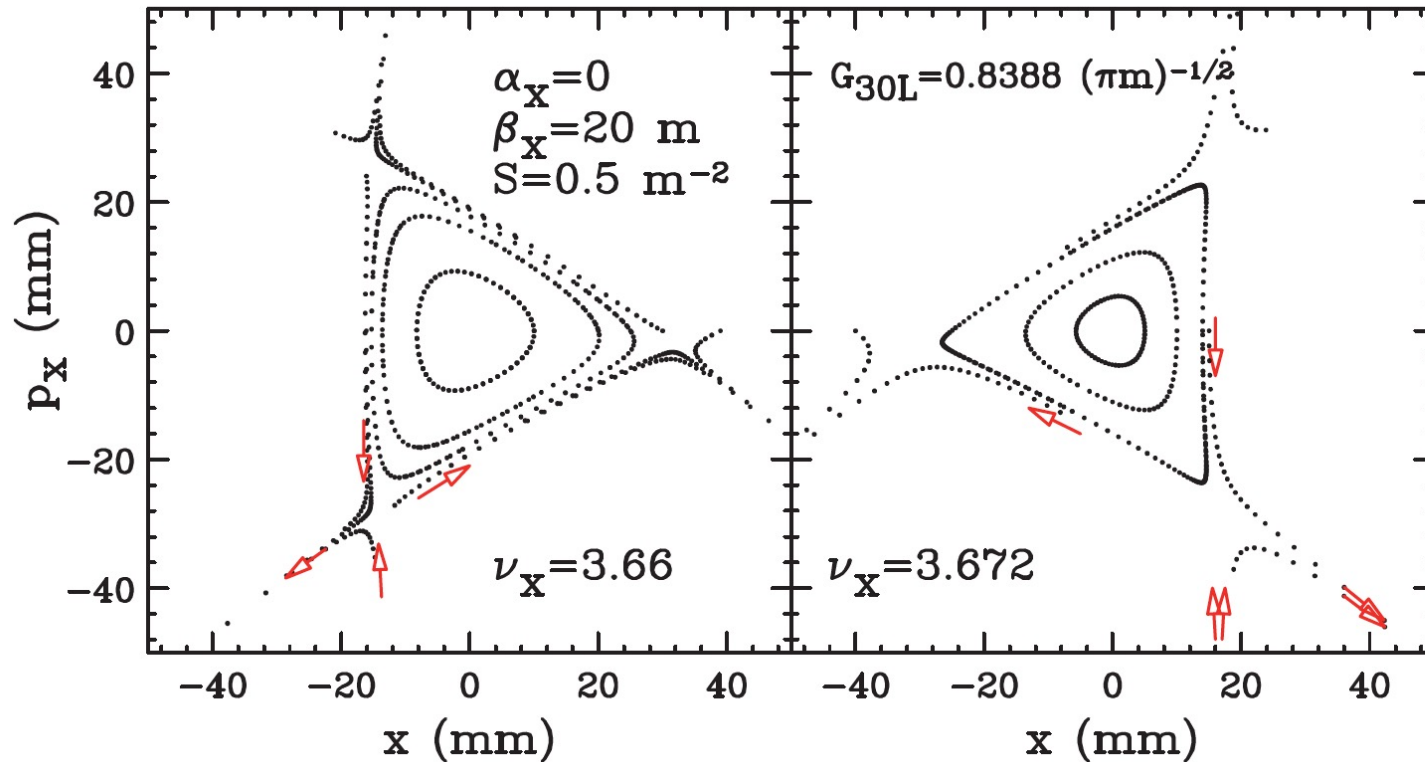
Poincaré map after adding a single sextupole



Poincaré surface of section of a particle near the 5th order resonances. Left without non-linear elements, right with one sextupole.

Figure from: <https://indico.cern.ch/event/509762/contributions/2711716/attachments/1520600/2880647/nlb.pdf>

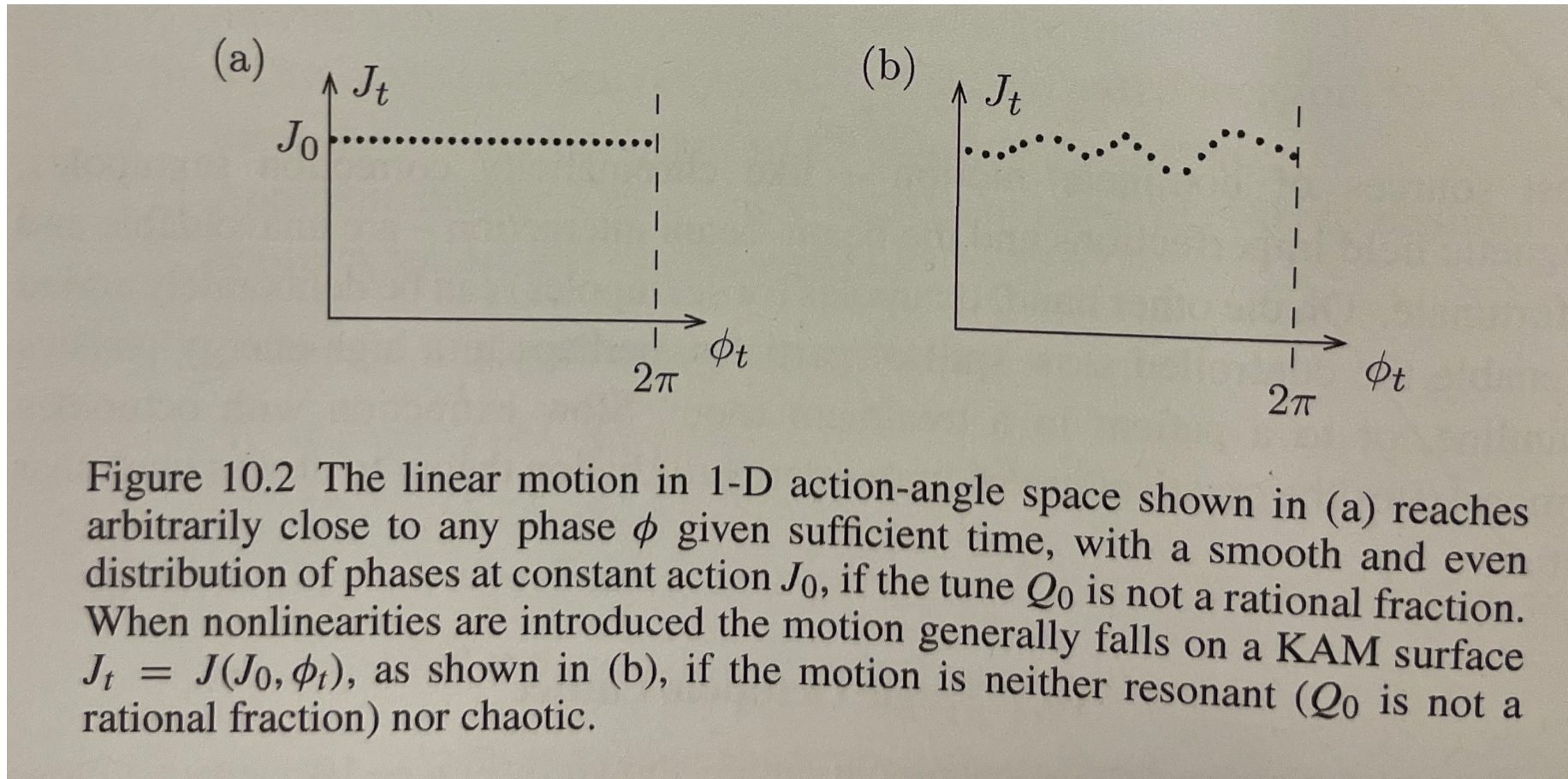
Poincaré map after adding a single sextupole



SY Lee

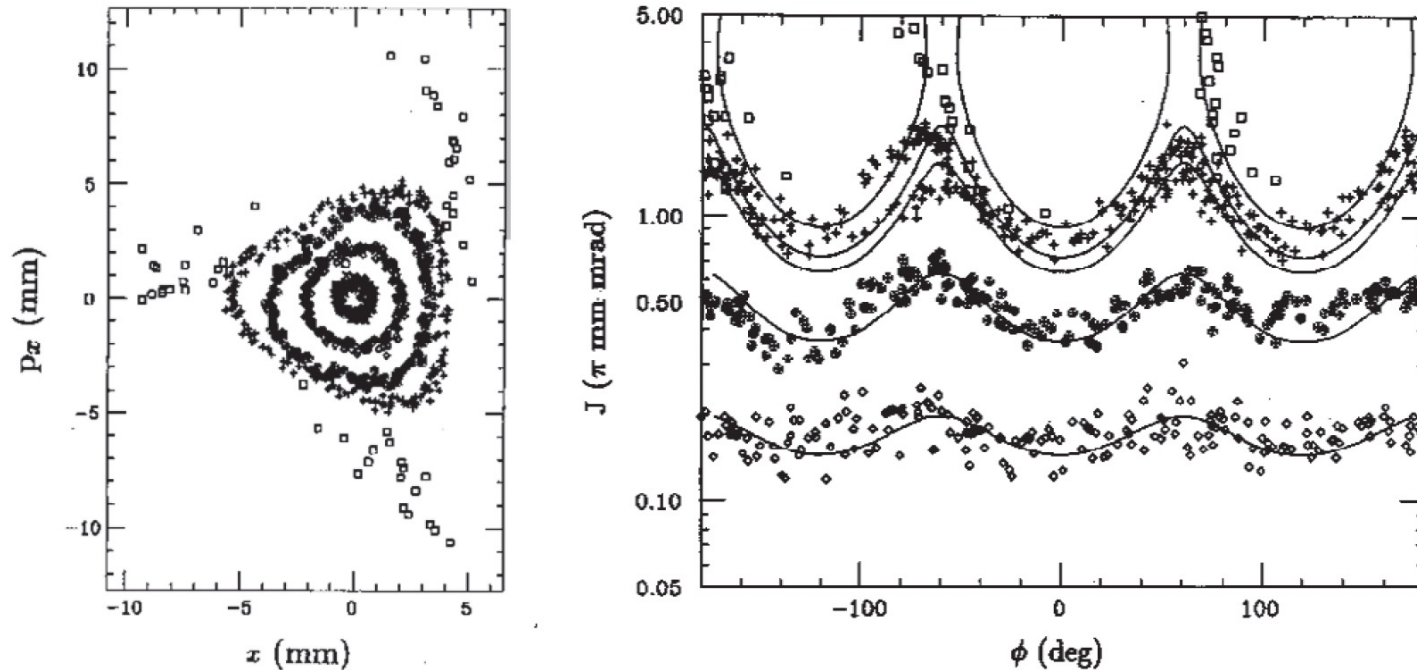
Figure 2.51: The Poincaré maps for betatron motion perturbed by a single sextupole magnet at a tune below (left) and above (right) a third order resonance. The integrated sextupole strength is $S = 0.5 \text{ m}^{-2}$ with lattice parameters $\beta_x = 20 \text{ m}$, and $\alpha_x = 0$. Arrows indicate directions of motion near a separatrix.

Another view: action-angle



From S. Peggs and T. Satogata, *Introduction to Accelerator Dynamics*

Another view: action-angle



SY Lee

Figure 2.53: Left: The measured Poincaré map of the normalized phase-space coordinates (x, p_x) of betatron motion near a third-order resonance $3\nu_x = 11$ at the IUCF cooler ring. Right: the Poincaré map in action-angle variables (J, ϕ) .

Another view: action-angle

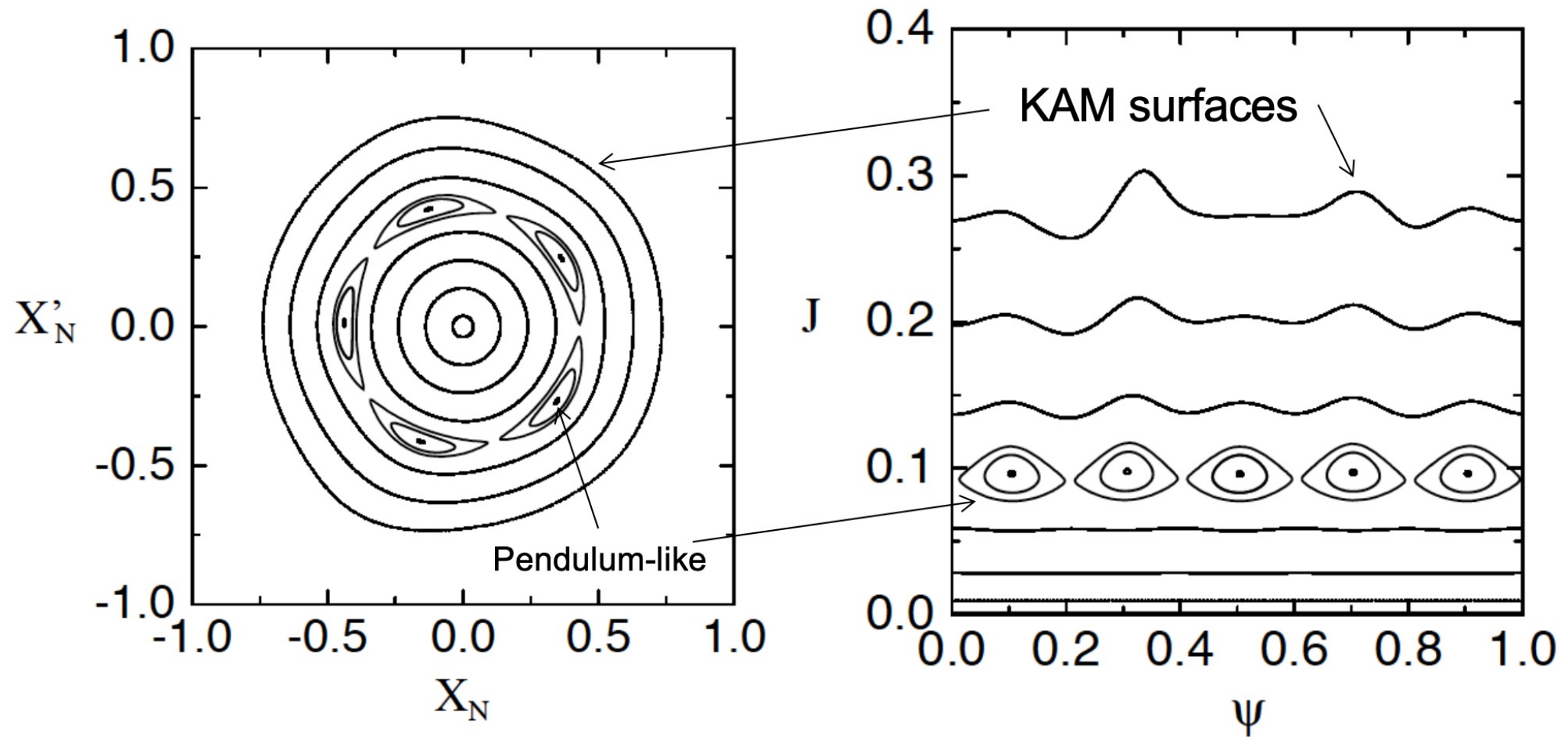
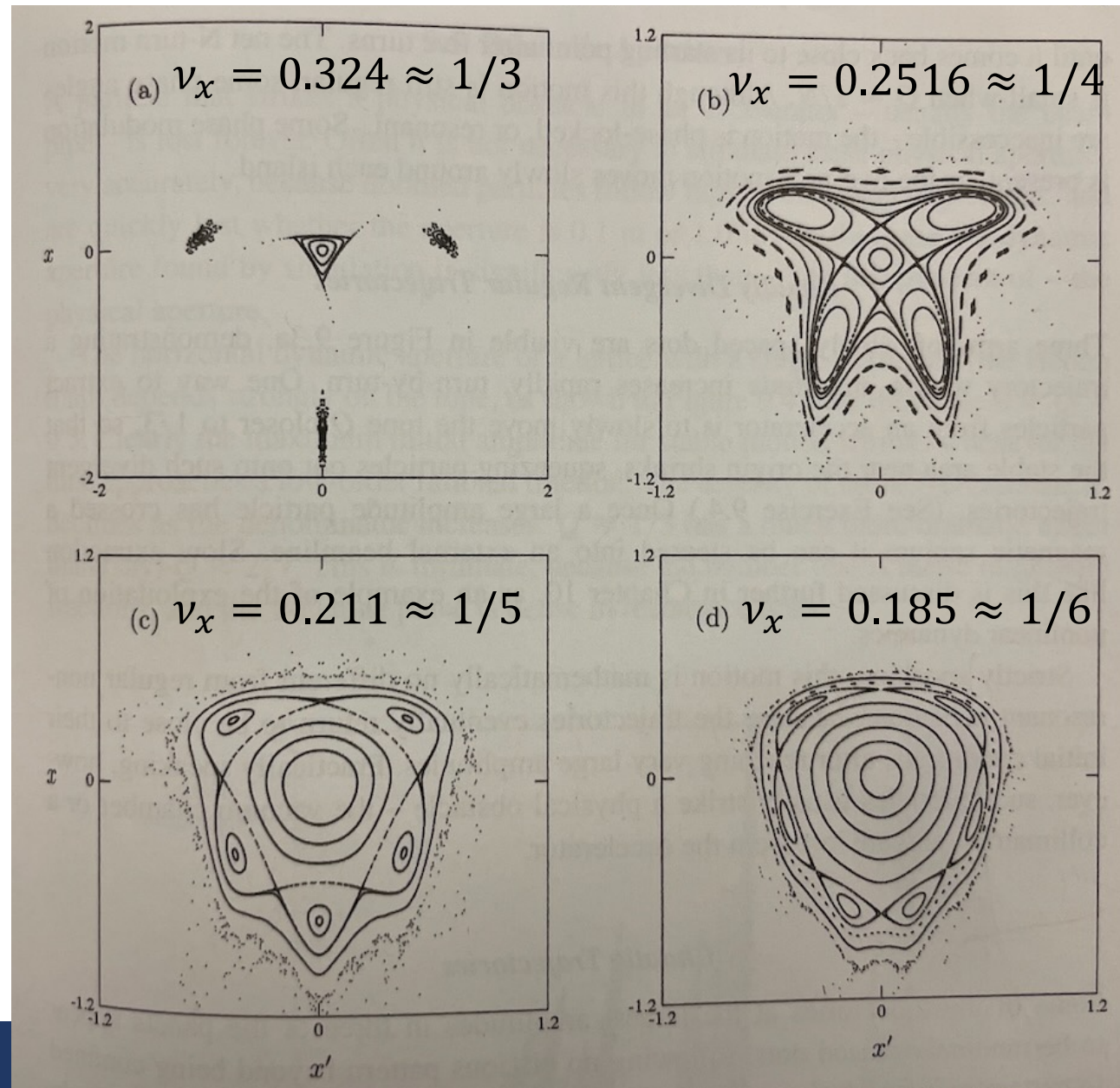


Figure from Todd Satogata, USPAS course: <https://toddsatogata.net/2019-USPAS/>

Examples at different tunes

From S. Peggs and T. Satogata,
Introduction to Accelerator Dynamics

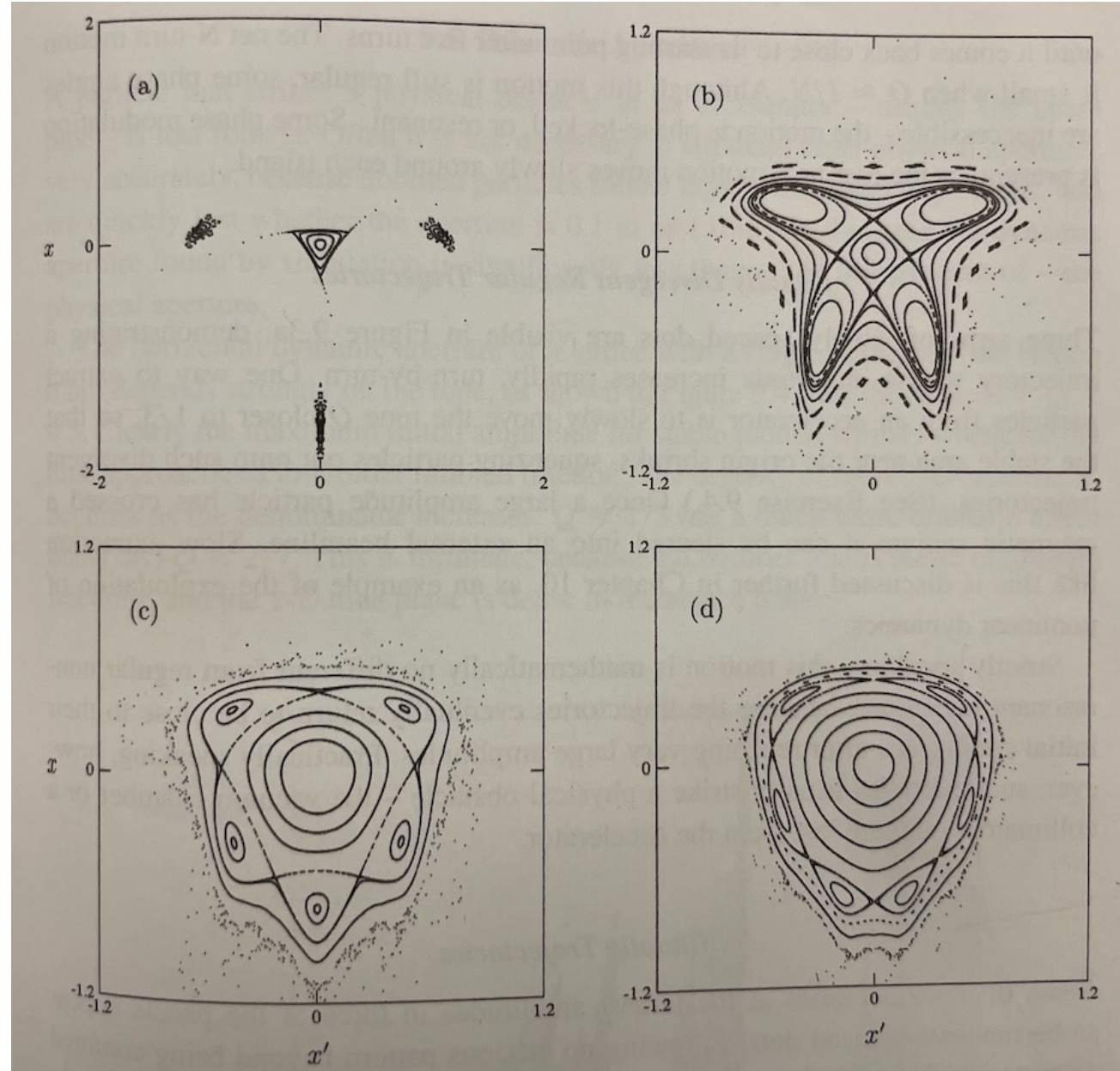
* Note swapped axes!



Types of trajectories

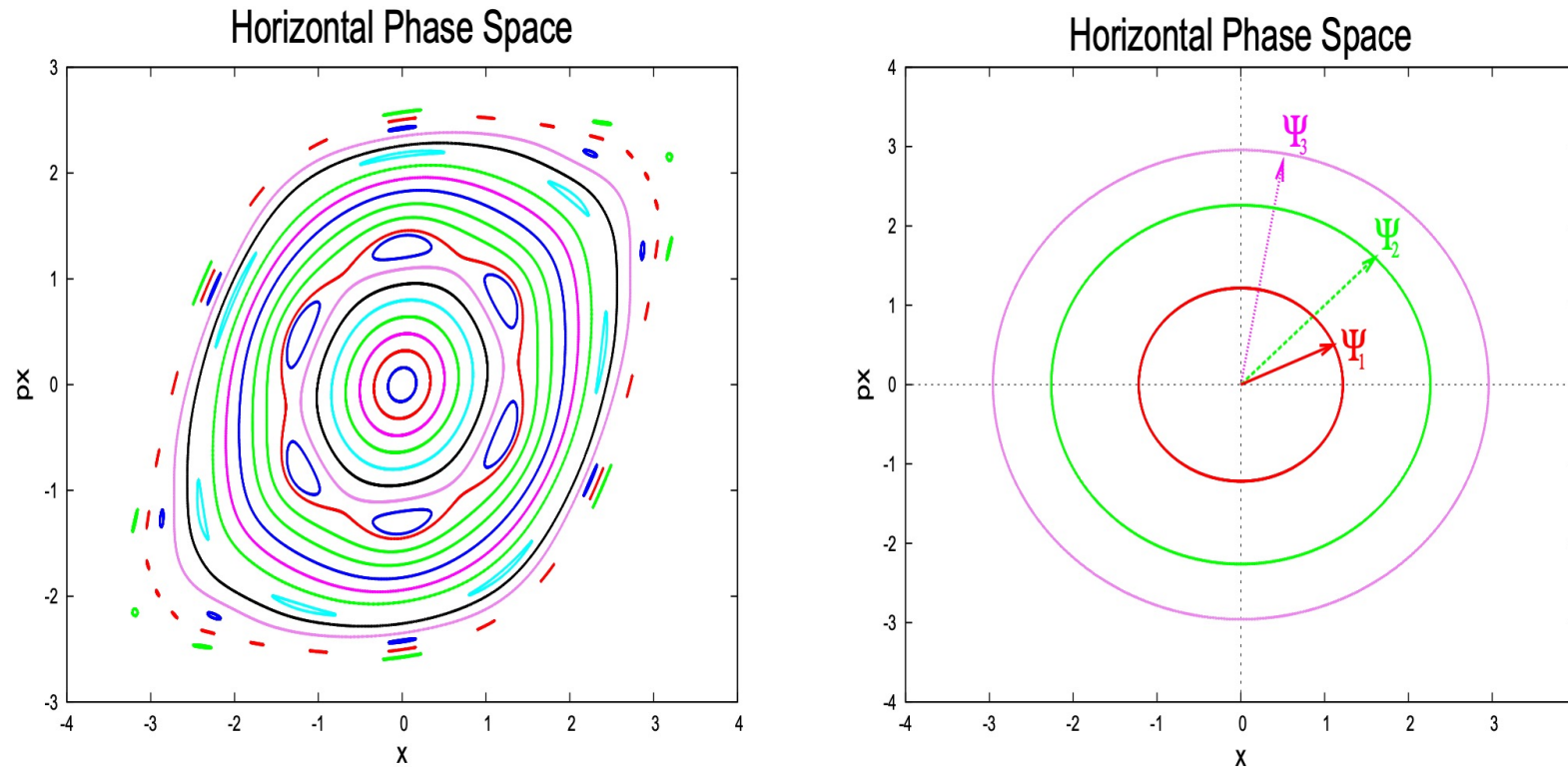
Adapted from From S. Peggs and T. Satogata,
Introduction to Accelerator Dynamics:

- Regular non-resonant: turn-by-turn trajectories fall on roughly circular, continuous lines near the origin
- Regular resonant: still follow closed curves, but non-circular distortions increase as amplitude increases; at high enough amplitudes, can hop from island to island each turn
- Rapidly divergent: trajectories are rapidly diverging lines, amplitude increases turn-by-turn; can be used to extract particles
- Chaotic: not well-defined trajectories; two nearby initial points in phase space will diverge rapidly



Evolution of phase space

Non-linearity makes the phase advance *amplitude-dependent*



Normal form transformation in the non-linear case, leading to amplitude dependent phase advance. The transformation was done for non-resonant amplitudes.

Figure from: <https://indico.cern.ch/event/509762/contributions/2711716/attachments/1520600/2880647/nlb.pdf>

Example for octupole

Normalised
displacement, x

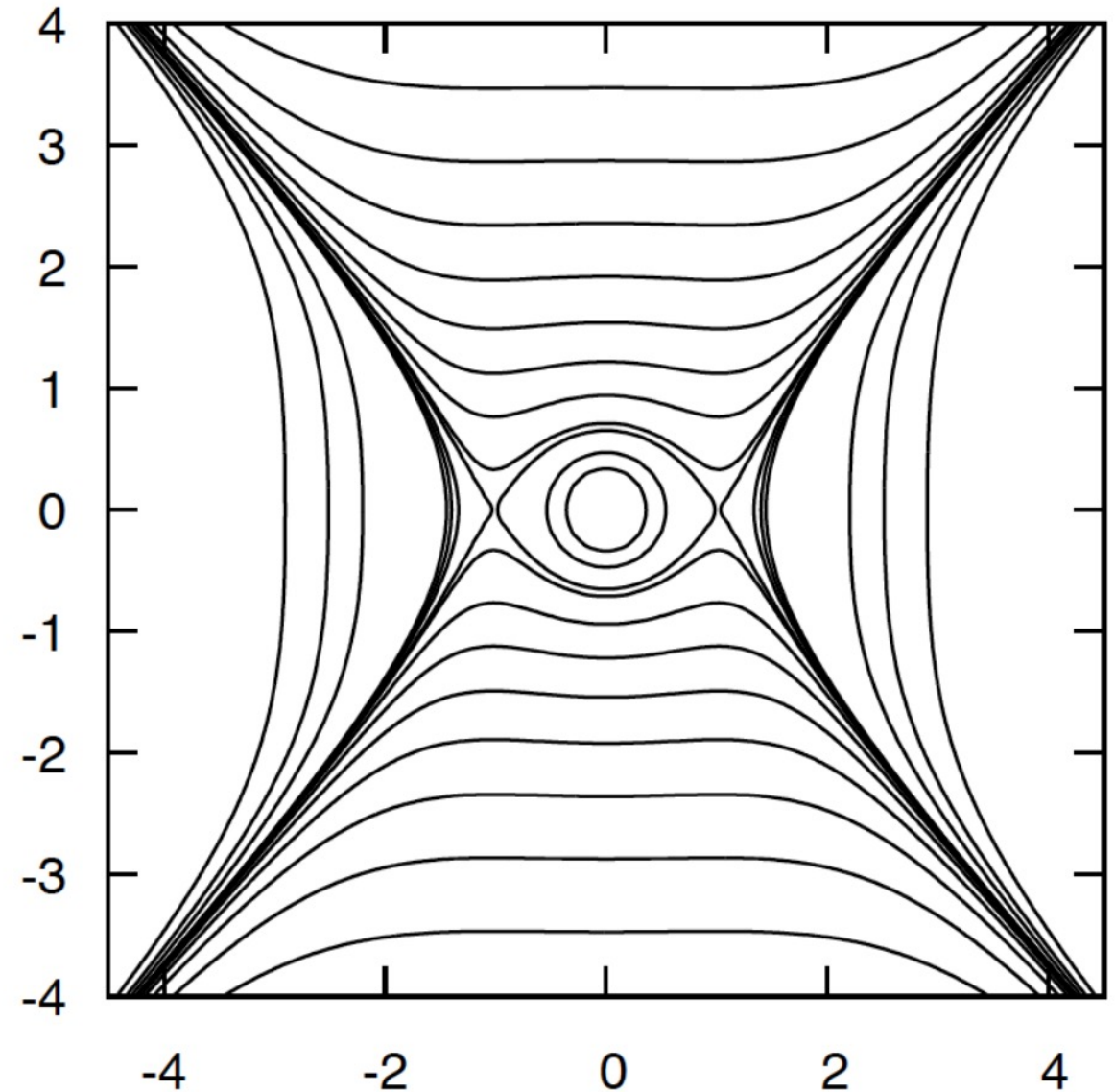
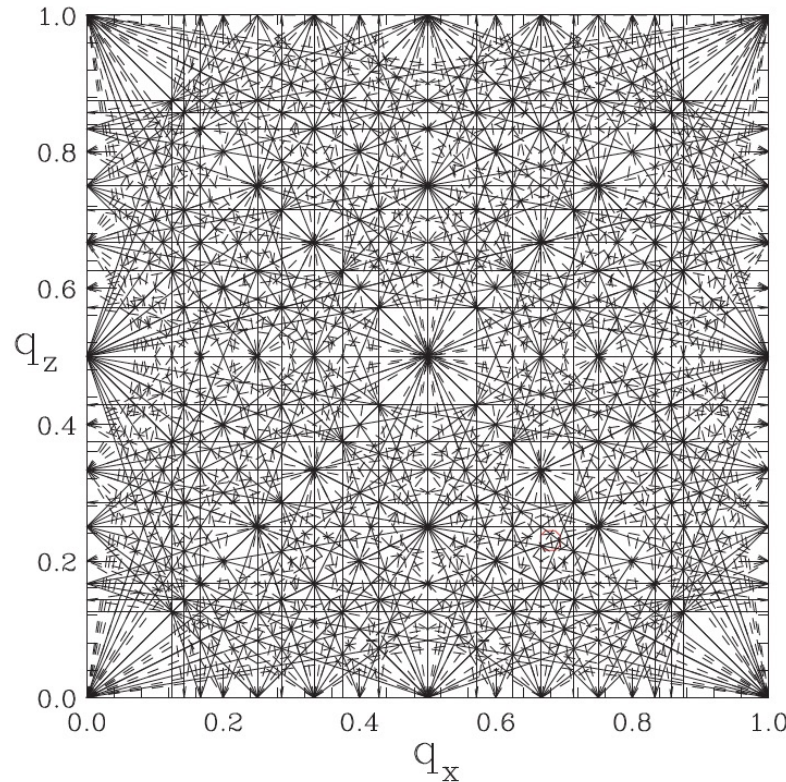
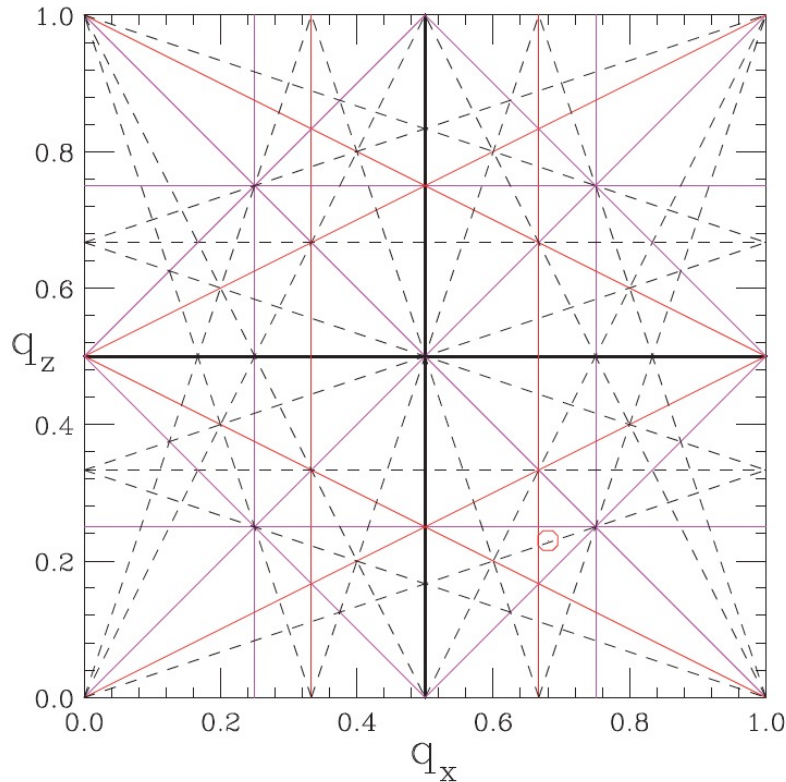


Figure from Todd Satogata, USPAS course:
<https://toddsatogata.net/2019-USPAS/>

Tune map with higher order resonances



SY Lee

Figure 2.57: The resonance lines: $m_x\nu_x + m_z\nu_z = \text{integer}$ with $|m_x| + |m_z| \leq 4$ (left) and $|m_x| + |m_z| \leq 8$ (Right). The symbol q_x and q_z are the fractional parts of betatron tunes ν_x and ν_z . When higher order resonances are important, resonance-free tune space becomes small.

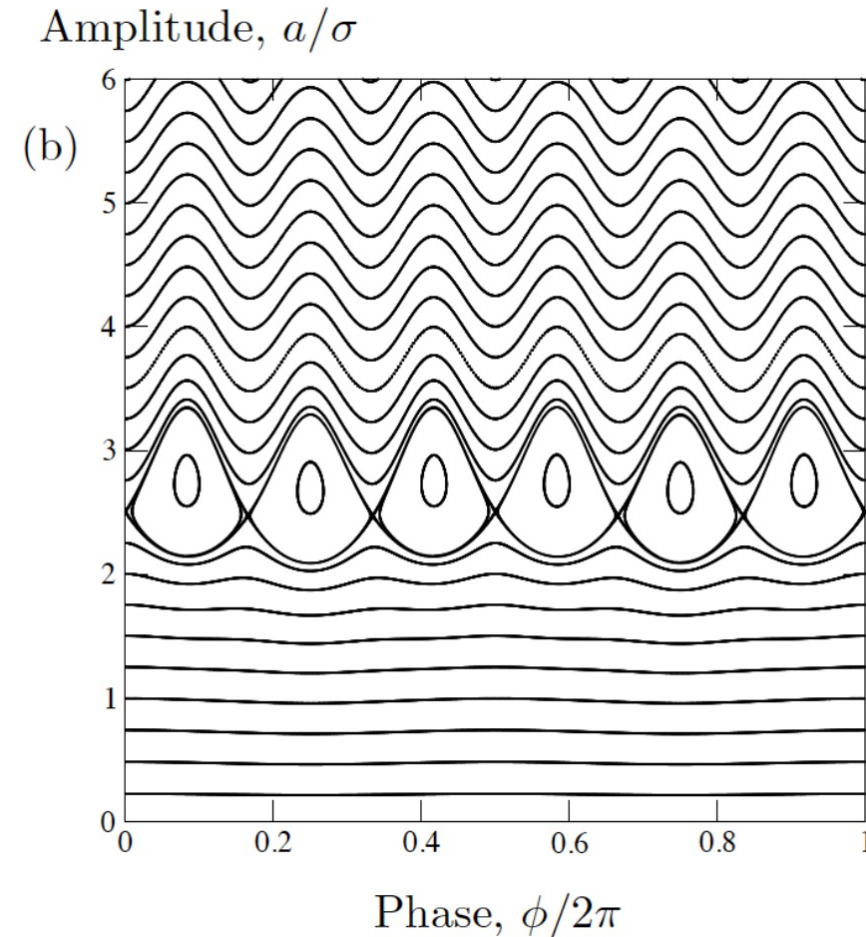
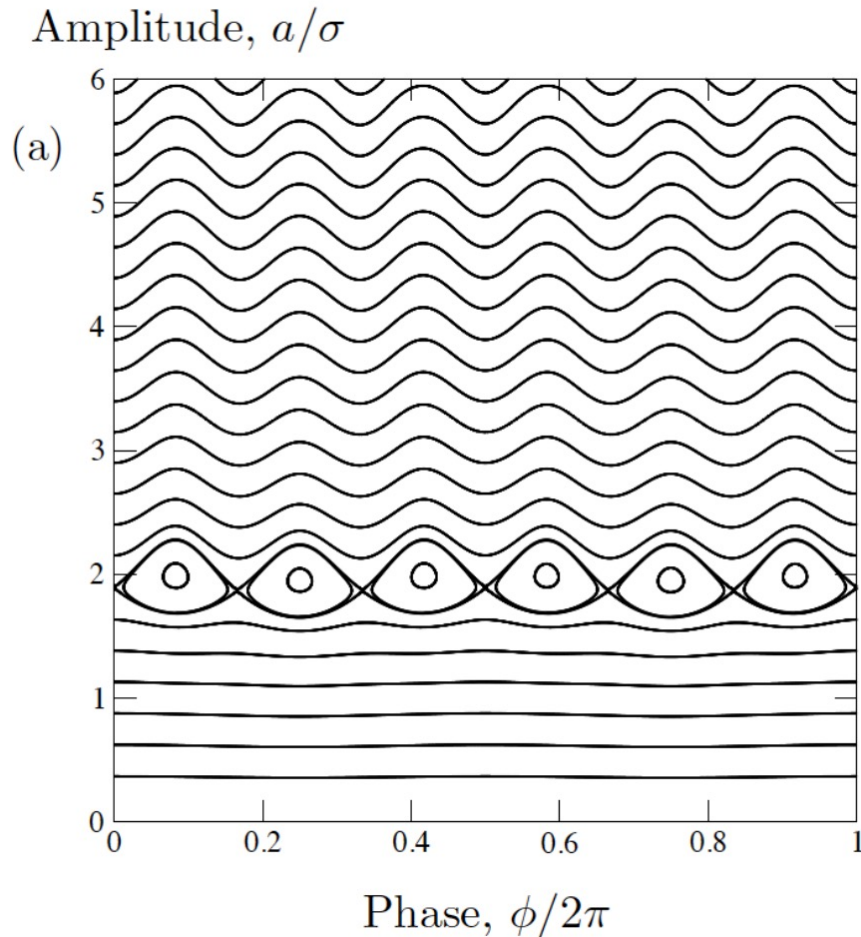
Bonus: spread from beam-beam interactions

In Peter's talk, he mentioned how beam-beam interactions increase the tune spread.

Shows up in the island amplitude widths:

Figure from Todd Satogata, USPAS course:

<https://toddsatogata.net/2019-USPAS/>



Ion sources

Overview of ion generation

- Wide variety of ion sources:
 - Method of creating ions
 - How method is employed with physical materials
- Some types:
 - Electron bombardment
 - Plasma discharge
 - Photon bombardment (laser sources)
 - Charge exchange reactions
- Could be an entire USPAS class; good set of reading resources:
<https://arxiv.org/html/1411.2445v1>

A note about reaction cross sections

- In atomic/molecular and nuclear physics, often use “cross section” σ to describe probability of reaction
- Called “cross section” because it has units of area
- Not *exactly* a probability, but we talk about it like it is one. Higher cross section means higher rate of reaction occurring.
- Can think about it effective area in a kinematic picture
- Common unit: “barns”

$$1 \text{ b} = 1 \times 10^{-28} \text{ m}^2 = 1 \times 10^{-24} \text{ cm}^2 = 100 \text{ fm}^2$$

- Cross sections depend on particle species, energies, underlying atomic, molecular, and nuclear physics
- Generally cross sections are measured: another thing we use accelerators for!

Electron bombardment

- Basic principle: electrons impact atom, can cause secondary electron emission, leaving behind an ion
- Coupled with some sort of electron gun and bias voltage to accelerate created ions
- Performance can be dominated by electron source performance

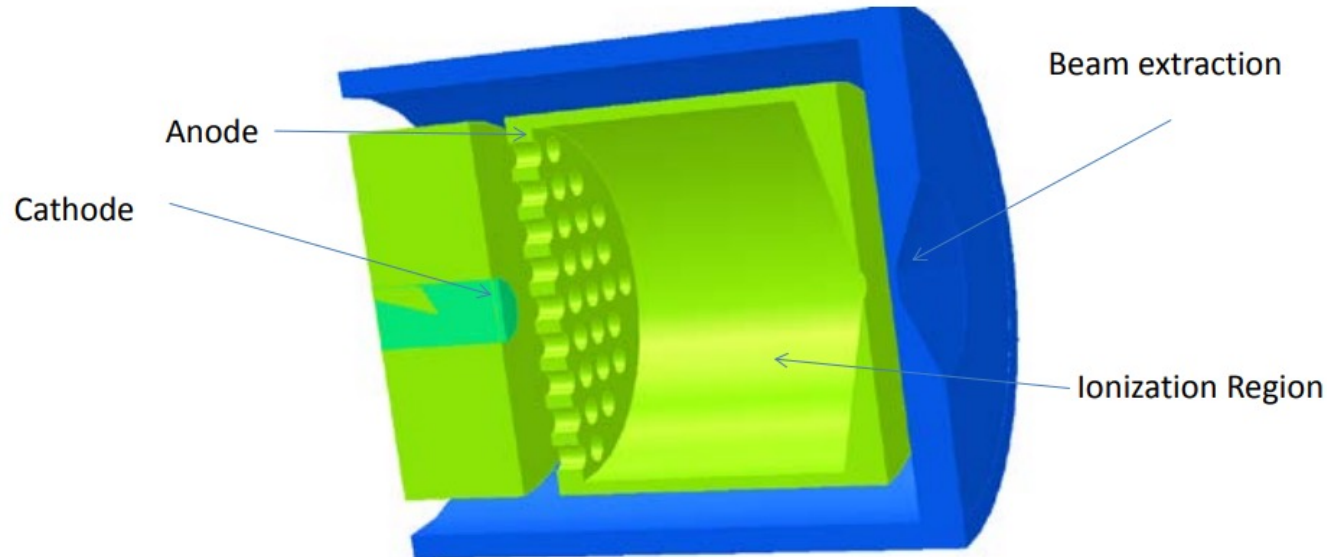


Fig. 3: Scheme of an electron impact ion source

<https://arxiv.org/pdf/1404.0918>

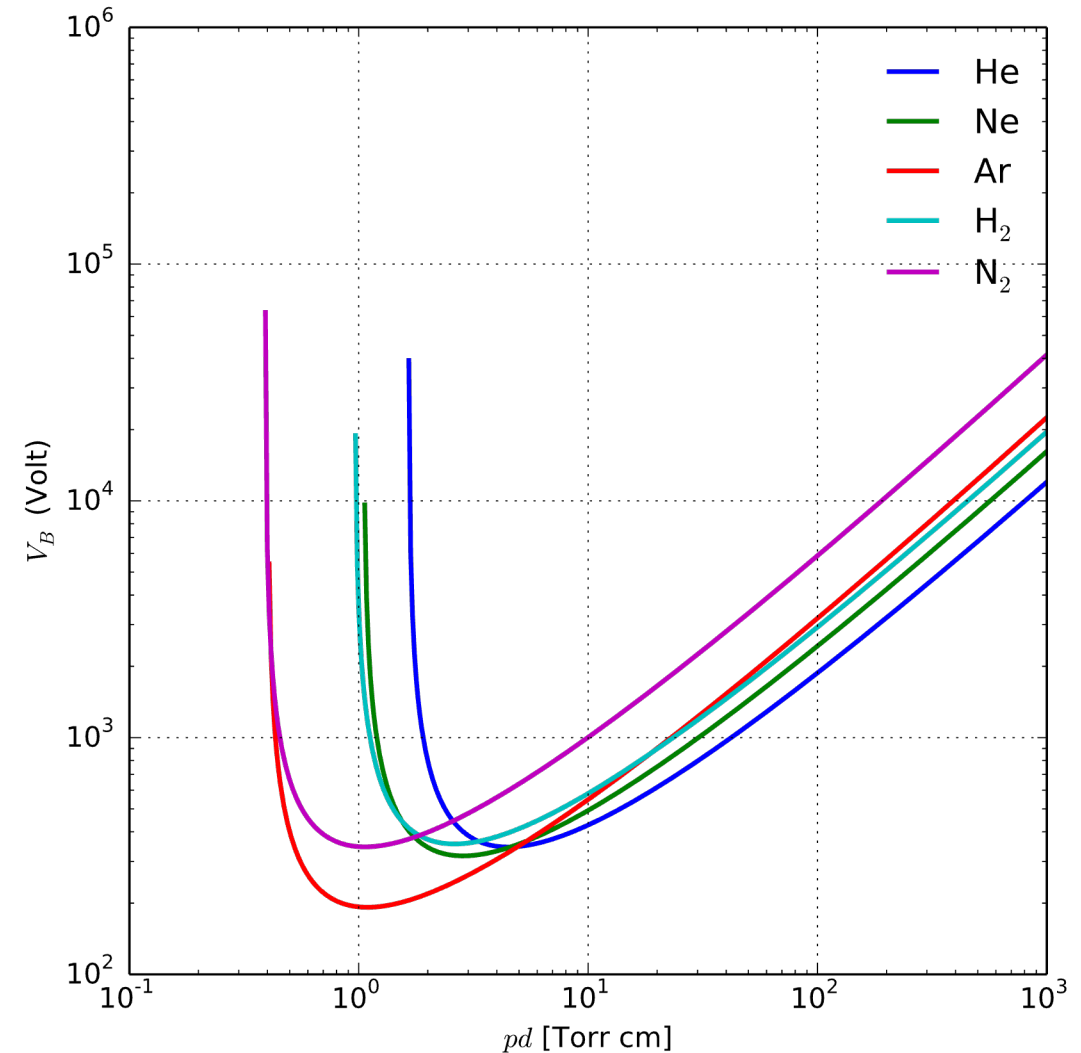
Plasma discharge – DC

- In high density neutral gas, intentionally cause breakdown with some DC voltage
- Electrons in plasma can go on to cause more ionization
- Voltage required for breakdown described by Paschen curves

$$V_b = \frac{aPd}{\ln(Pd) + b}$$

P Pressure
 d Electrode distance
 a, b Gas constants

<https://arxiv.org/pdf/1404.0918>



Plasma discharge

- Can add a magnetic field for confinement or steering
- Path lengths/rate of collisions described by plasma physics
- Can also involve sputtering of atoms off of cathode electrode surface
- Field used for breakdown can be RF field instead of DC voltage

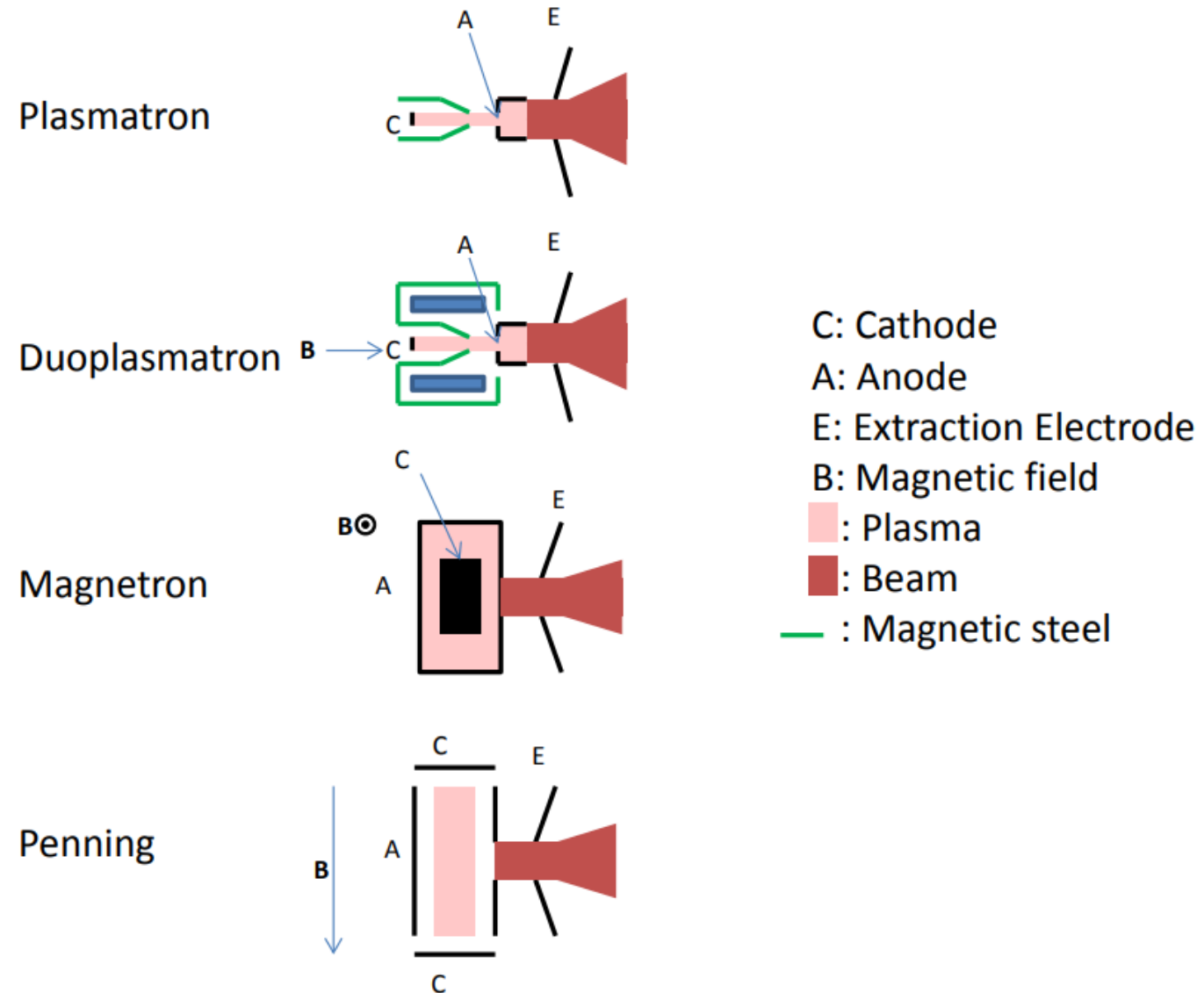


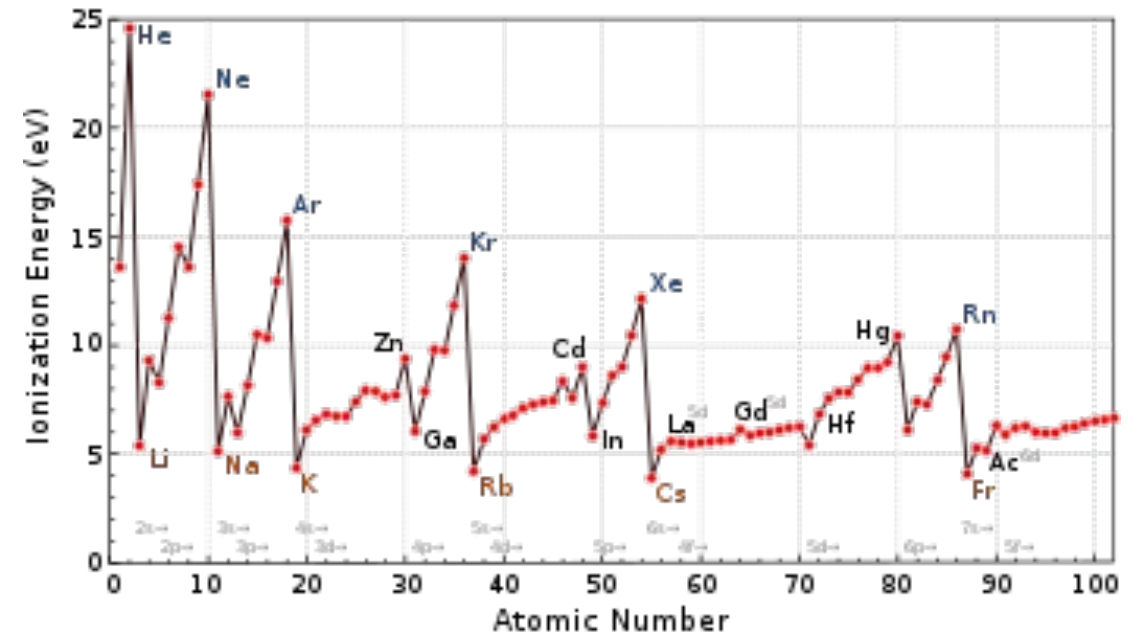
Fig. 8: Schematic configuration of a few discharge-type ion sources

Related: RF using electron-cyclotron resonance

- Apply some magnetic field, electrons will oscillate with cyclotron frequency
- Can heat electrons with input RF at this frequency
- Electrons go on to ionize neutrals more effectively

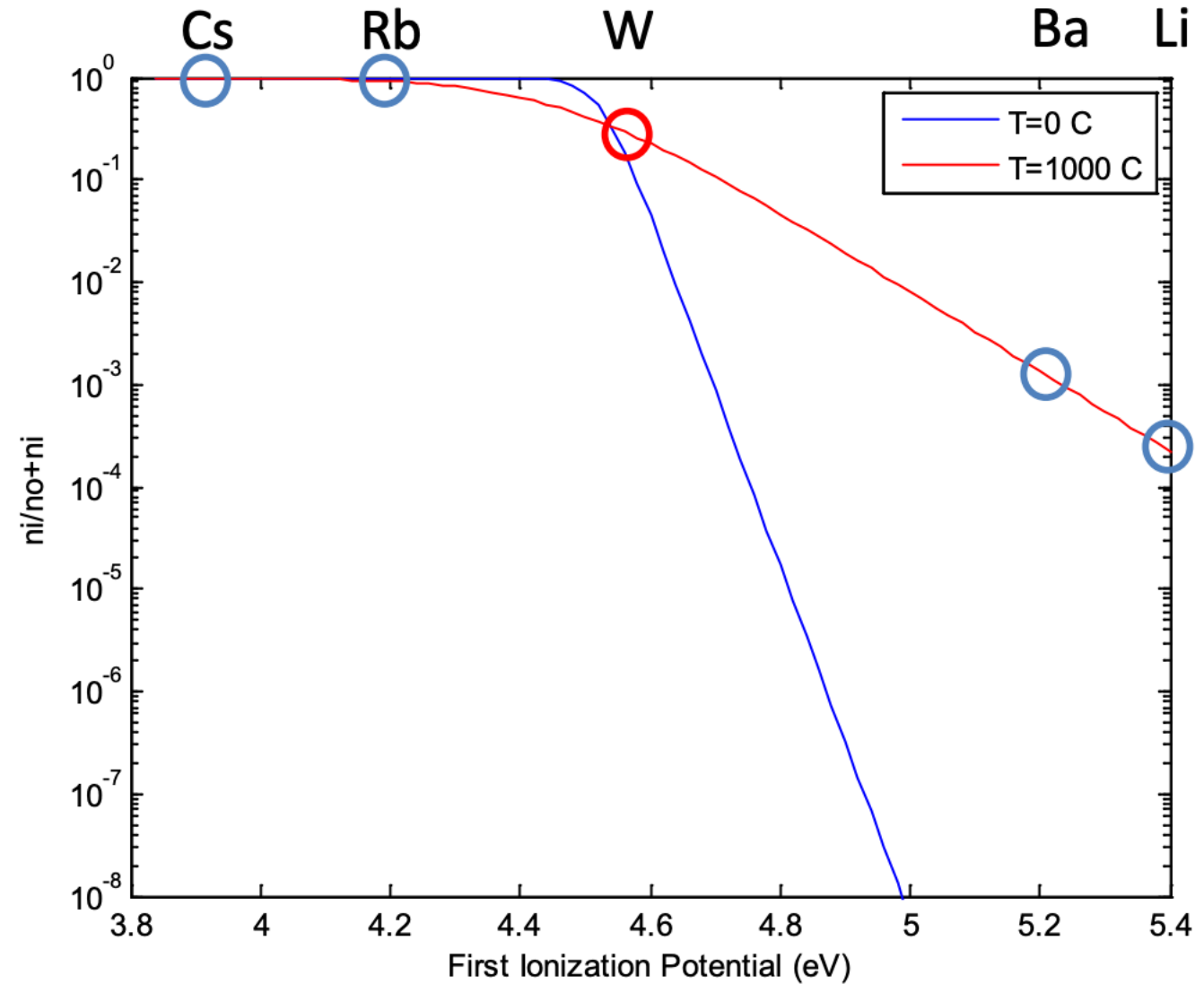
Photon bombardment

- Incident photons can supply enough energy to electrons to excite them/ionize neutral particle
- Choice of energies (frequencies) depends on energy levels
- Ionization energies are on the order of ~ 10 eV
 - Requires UV to EUV (extreme ultraviolet) lasers OR
 - Requires initial excitation with lower energy laser, followed by additional second excitation



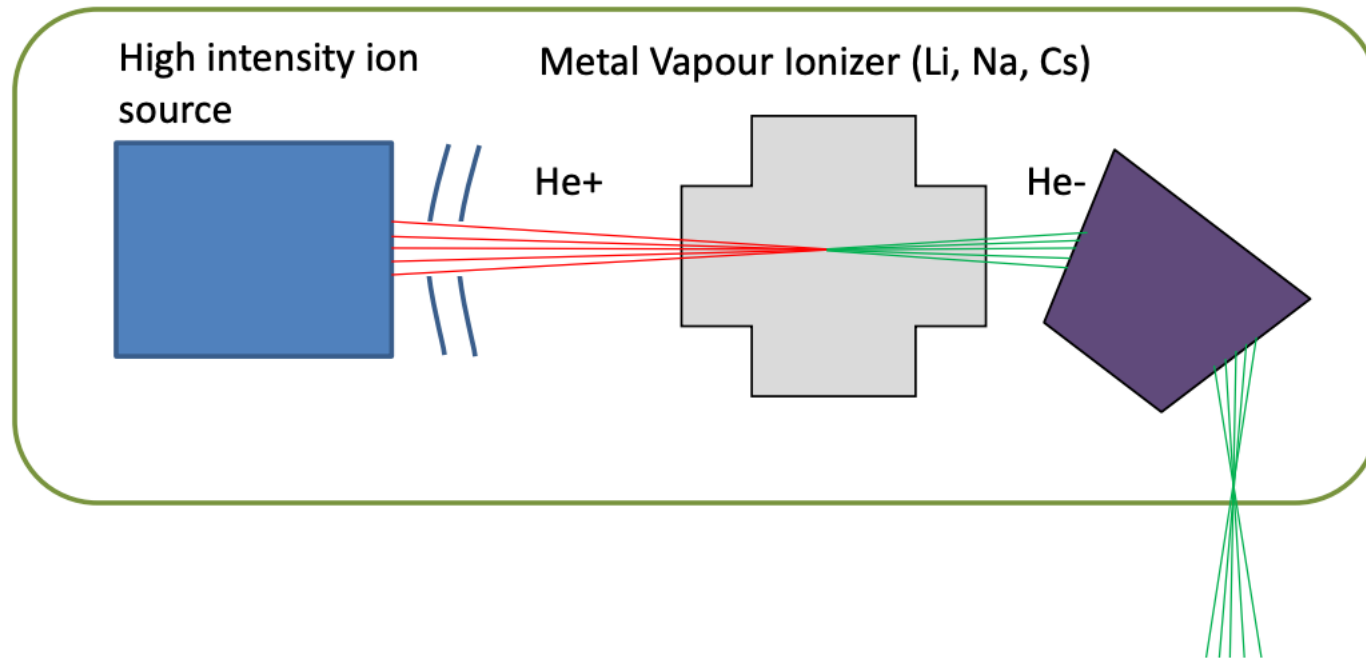
Surface desorption

- Similar starting principle to thermionic cathode:
 - Surface is heated, some electrons will be freed
 - Surface atoms liberated (could be a variety of methods)
 - Some fraction of those atoms will already be ionized
- Plot: fraction of atoms desorbed as ions vs total (neutrals + ions)



Charge exchange

- Can start with positively charged ions and neutral atom to get negatively charged ions or vice versa
- Cross sections depend on electron affinity of specific atoms
- Example: H^- production using H^+ and neutral hydrogen (could be other neutral)
- Can also be a limiting factor which reduces overall ionization fraction for other kinds of sources



Ionizing radiation

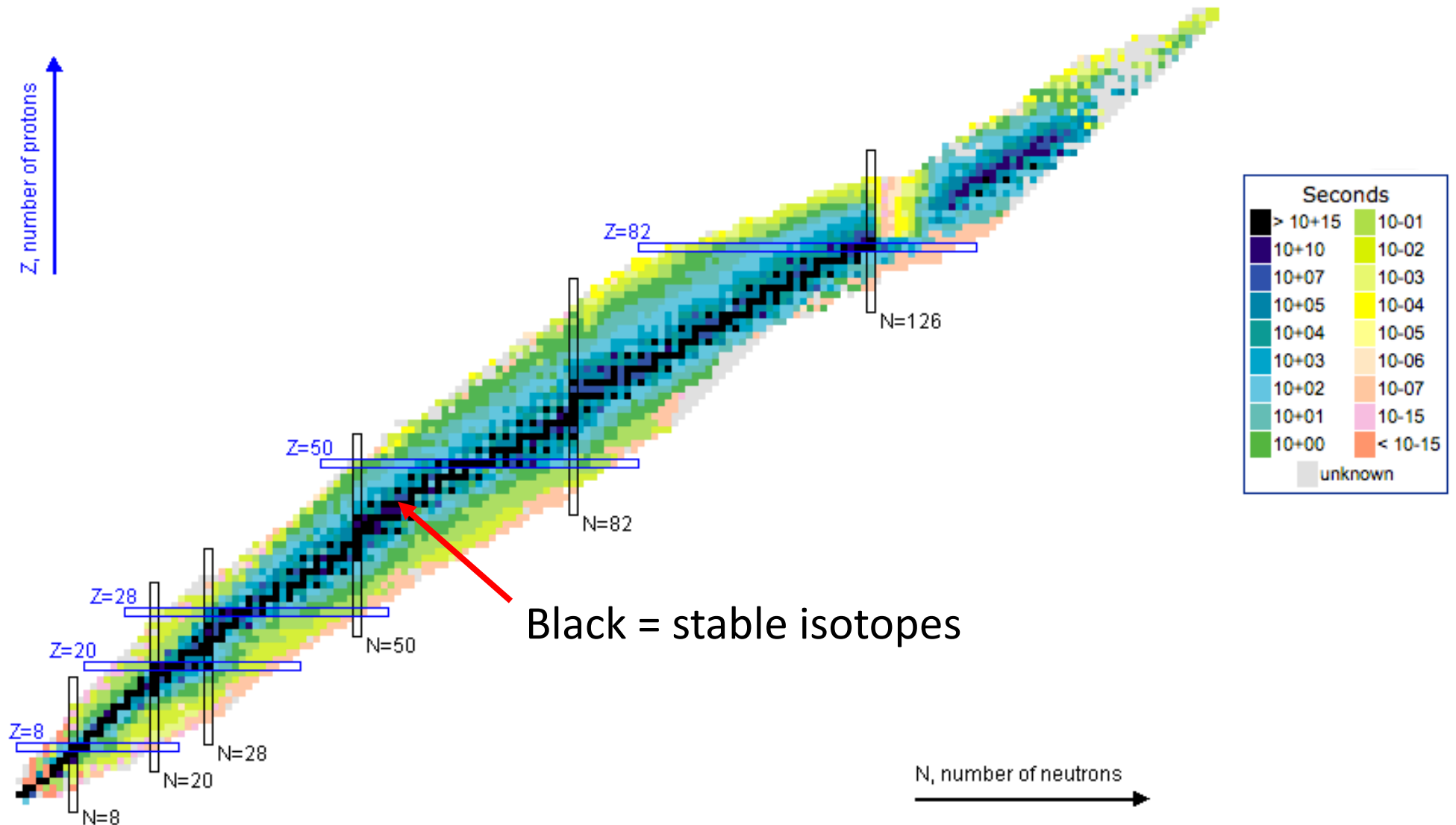
Ionizing radiation

- Radiation is categorized as “ionizing” or “non-ionizing”
- For radiation damage, we are concerned with ionizing radiation
- Types of ionizing radiation:
 - Photons (X-rays and gamma rays)
 - Electrons
 - Protons
 - Alpha particles (He-4 nuclei)
 - Neutrons
 - Heavy ions
- Full radiation damage picture includes not just incident initial radiation, but products from the radiation

Sources of radiation

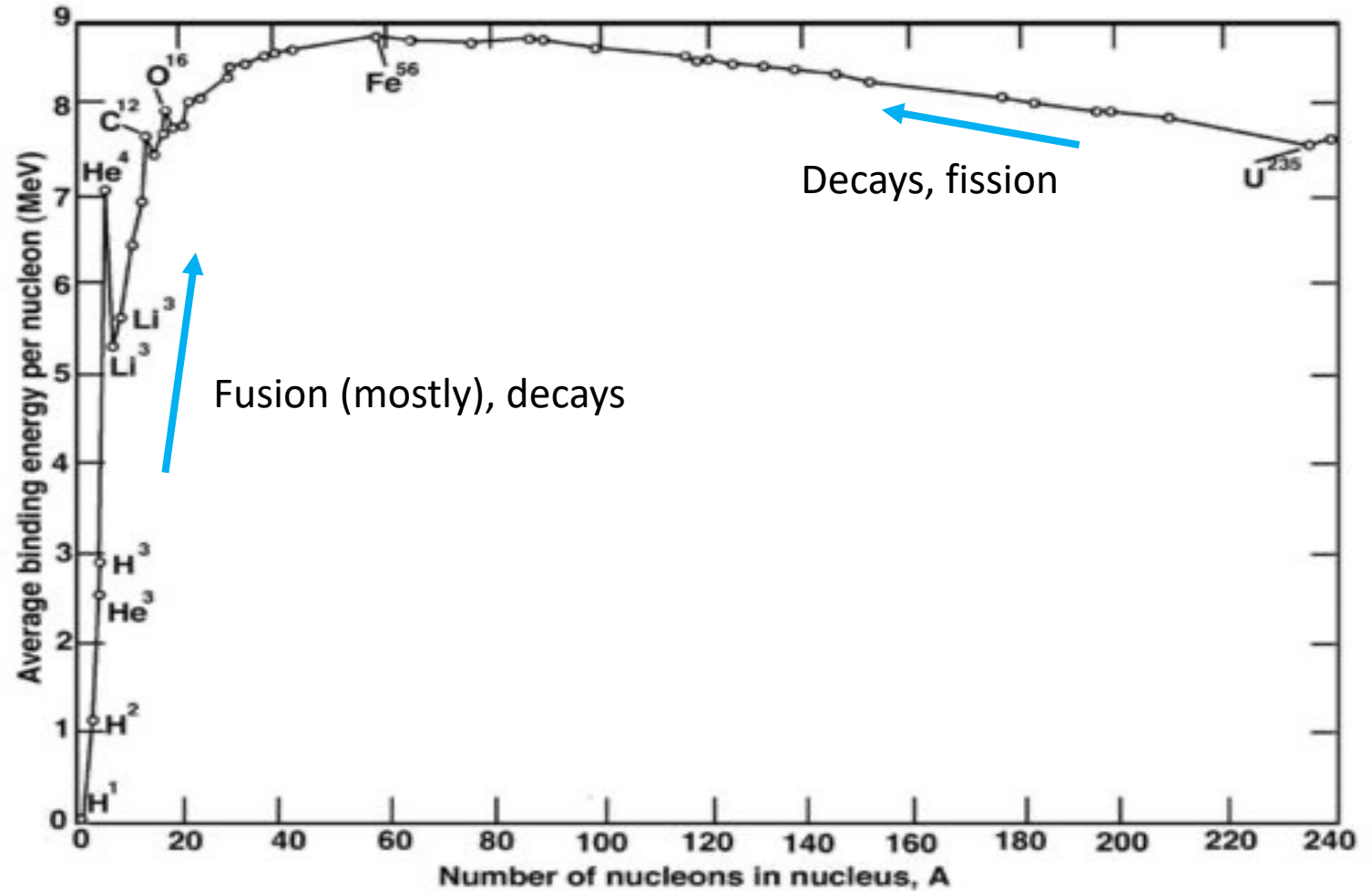
- In general:
 - Radioactive decay of unstable nuclei
 - Energy level transitions in atoms (X-rays) and nuclei (gamma rays)
 - Nuclear reactions
- In accelerators:
 - Beam particles themselves
 - Radiation emitted from the beam
 - Both interacting with targets, other material
 - Transmutation products

Chart of the nuclides: map of isotope stability

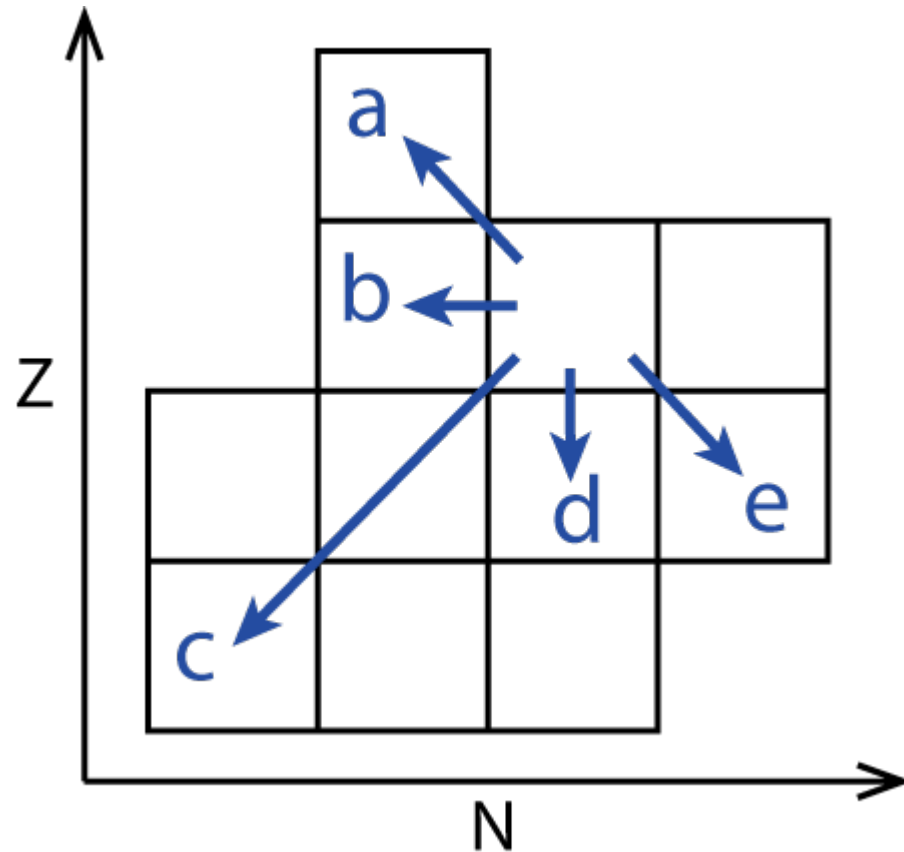


Source of instability

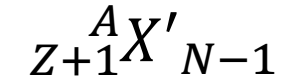
- Isotopes decay to minimize energy
- Move towards HIGHER binding energy per nucleon
- Decays occur because they're energetically favorable
- What is energetically favorable is determined by nuclear physics (structure of energy levels)



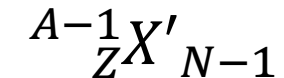
Decay modes – between isotopes



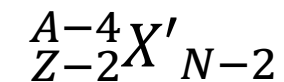
a. Beta minus (β^-)/electron



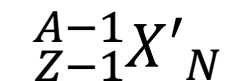
b. Neutron emission (n)



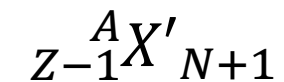
c. Alpha (α)



d. Proton emission (p)

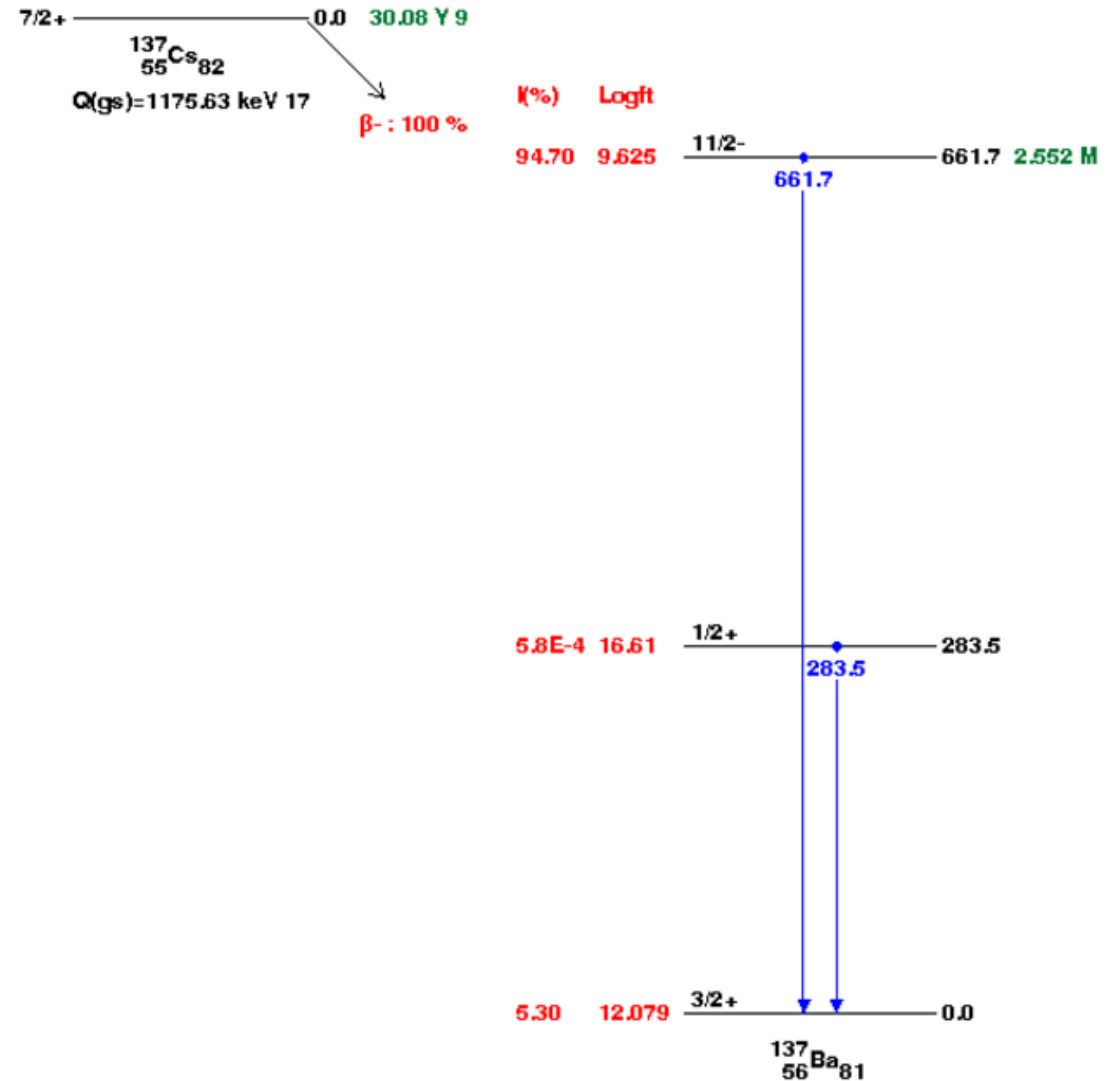


e. Beta plus (β^+)/positron AND
electron capture (ϵ)

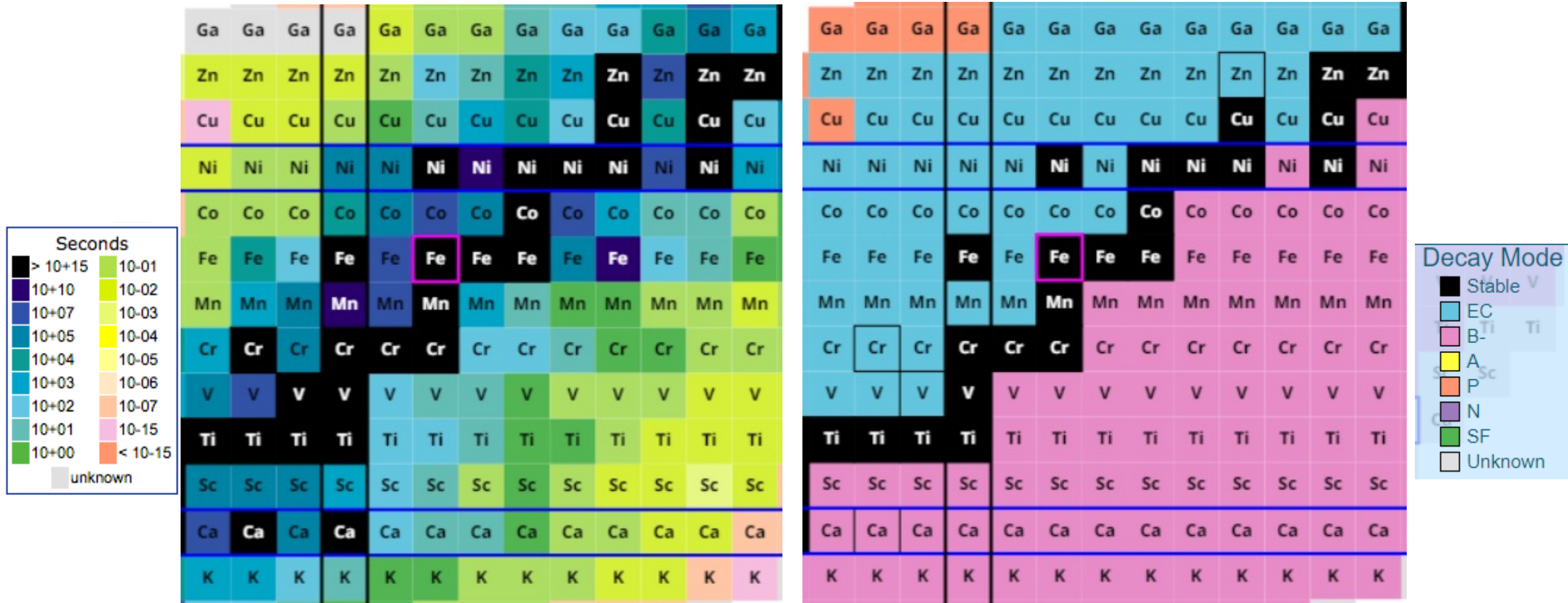


Decay modes – between energy levels

- Decay of isotope in excited state from one nuclear energy level to another produces gamma ray
- Decay of other isotopes can produce stable isotope in an excited state



Example: region around Fe-56



Half life and lifetime

- Decay is described statistically by the decay rate λ

$$N(t) = N_0 e^{-\lambda t}$$

$$\lambda = -\frac{dN/dt}{N}$$

Half-life:

Amount of time it takes for half of the nuclei present to decay

$$t_{1/2} = \frac{\ln(2)}{\lambda} \approx \frac{0.693}{\lambda}$$

(Mean) lifetime:

Average amount of time a nucleus is likely to survive before it decays

$$\tau = \frac{\int_0^\infty t |dN/dt| dt}{\int_0^\infty |dN/dt| dt} = \frac{1}{\lambda}$$

Activity definition and units

- Measure of radioactive decay: “activity”

$$A(t) = \lambda N(t) = A_0 e^{-\lambda t}$$

- Historical/common units of “Curies” Ci
- SI unit of “Becquerels” Bq

$$1 \text{ Ci} = 3.7 \times 10^7 \text{ decays/second}$$

$$1 \text{ Bq} = 1 \text{ decay/second}$$

- Only quantifies number of decays, not energy, type, or potential impact

Radiation exposure

- Exposure quantifies amount of ionization produced in a material from gammas or X-rays passing through air (charge per mass)

$$X = \frac{Q}{m}$$

- SI units: Coulombs/kg; common unit: “Roentgens” R
- Most nuclei can/do emit more than one gamma
- Exposure also depends on distance from source
- Exposure rate includes these:

$$\frac{\Delta X}{\Delta t} = \Gamma \frac{A}{d^2}$$


Nuclide-specific constant, related to intensity/energy of gammas

Dose definitions and units

- Roentgens are defined for ionization in air, other materials will have different amounts of absorbed energy/ionization
- This is quantified by “absorbed dose” D in units of rad (radiation absorbed dose)
 - $1 \text{ rad} = 0.88 \text{ R}$
 - SI unit is the “Gray”, $1 \text{ Gy} = 100 \text{ rad}$
- Effects of different kinds of radiation on the human body vary
- Attempts to capture this: relative biological effectiveness (RBE) and quality factor (QF)
 - RBE: ratio of dose to amount of X-rays that produces the same biological effect, hard to measure
 - QF: specific to type and energy of radiation, quantifies energy deposited for unit path length

Dose equivalent

- QF for different types of radiation

Radiation	QF
X-rays, β , γ	1
Low energy p, n ($\sim$ keV)	2-5
Energetic p, n ($\sim$ MeV)	5-10
α	20

- Dose equivalent is the dose we use for humans:

$$DE = QF \cdot D$$

- Units:
 - rem (roentgen equivalent man)
 - “Sievert”, 1 Sv = 100 rem

Summary of radiation units

- Different quantities related to radiation

Quantity	Measure of	Traditional unit	SI unit
Activity (A)	Decay rate	Curie (Ci)	Becquerel (Bq)
Exposure (X)	Ionization in air	Roentgen (R)	Coulomb/kg
Absorbed dose (D)	Energy absorption	rad	Gray (Gy)
Effective dose (DE)	Biological effectiveness	rem	Sievert (Sv)

- Some dose numbers:
 - Radiation worker limit per year from NRC = 50 mSv (5 rem)
 - Yearly background radiation = 4 mSv (400 mrem)
 - Chest CT scan = 7 mSv (700 mrem)
 - Dental X-ray = 5 μ Sv (0.5 mrem)

Radiation interactions with matter

Types of interactions

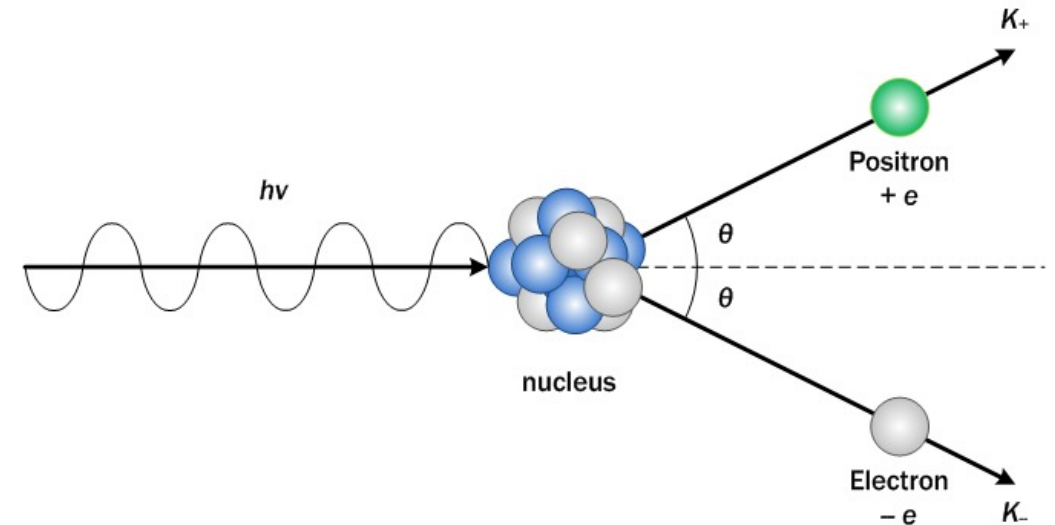
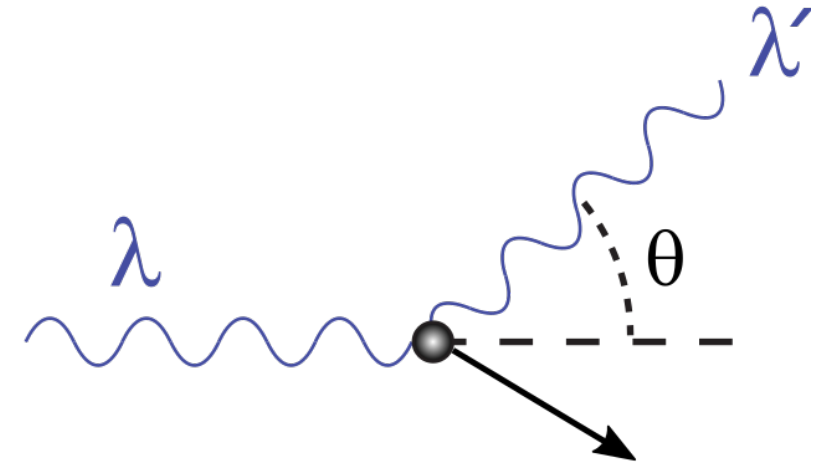
- Two overall categories: nuclear and electronic
 - Electronic interactions are important for charged particles and photons
 - Nuclear interactions can occur for all types of radiation
 - Electronic interactions tend to dominate overall
- There are many sub-categories to these two overall types
- Not every interaction produces ionization directly
- Which interactions are dominant depends on the particle species, its energy, and the material it is interacting with

Conservation laws/kinematics

- Many quantities are conserved: total energy, linear momentum, angular momentum, baryon number, lepton number
- Scattering reactions look a lot like normal kinematics: more energy transferred for similarly-sized incident and target particles
 - E.g. two billiard balls vs a golf ball and a bowling ball
 - Consider both elastic and inelastic scattering
 - Reactions can be many-body problems
- *Total* energy is important
 - Includes kinetic energy, rest mass energy, and binding energy
 - Reactions can be endo- or exothermic (requiring energy or releasing energy)

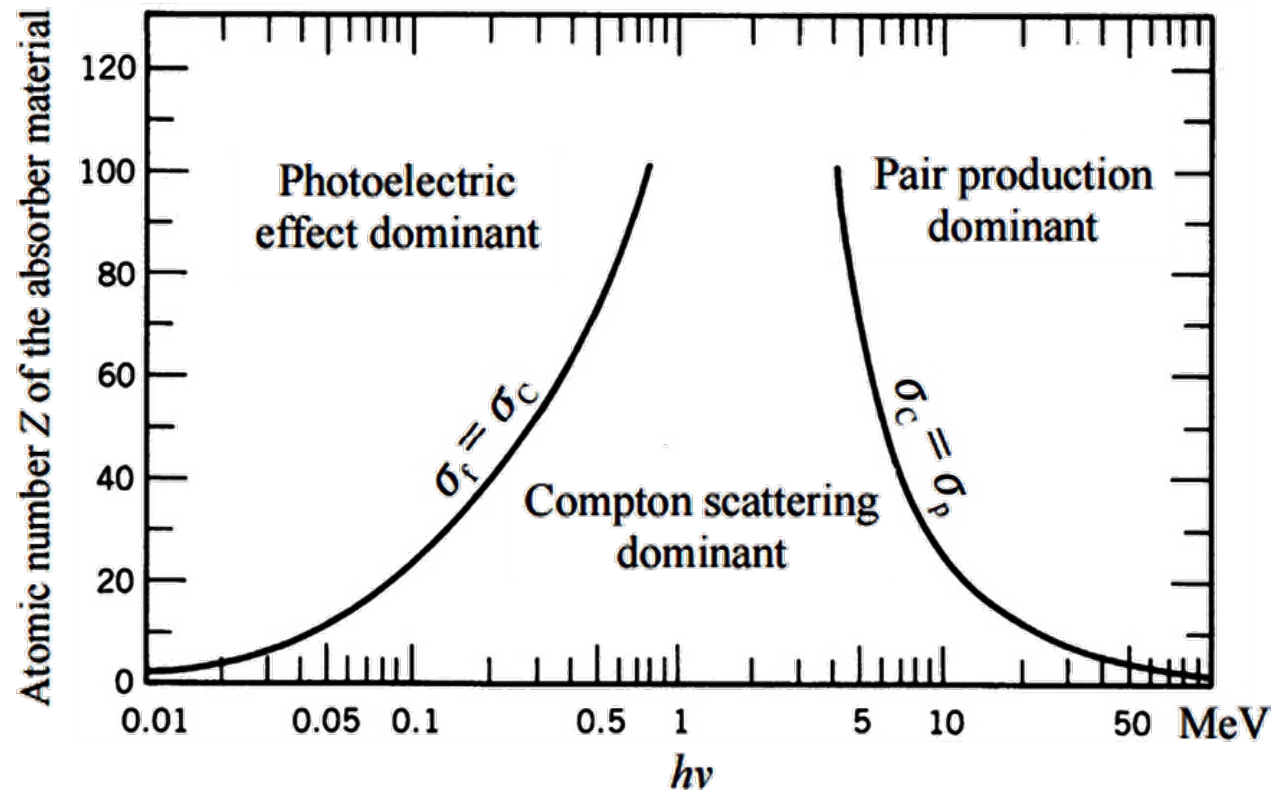
Photon interactions

- Photoelectric absorption
 - Photoelectric effect: photon is absorbed by atom, electron is released
 - Photon is gone
- Compton scattering
 - Scattering of photon from (mostly) weakly bound electrons
 - Photon energy is reduced
- Pair production
 - Photon becomes electron-positron pair
 - Has threshold energy of $2m_e$
 - Requires nearby nucleus, very small recoil conserves momentum



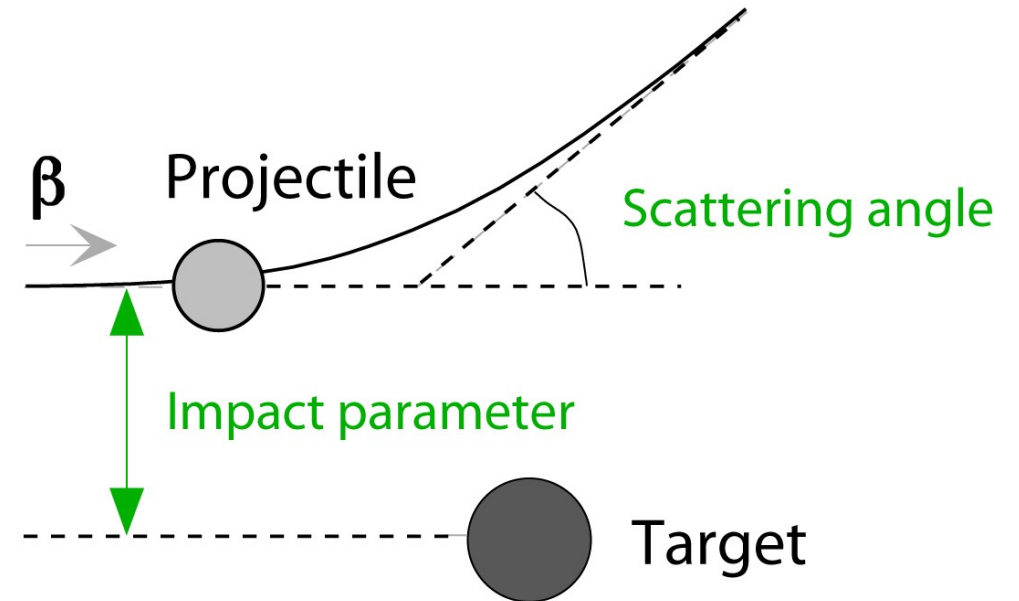
Photon interactions

- Photoelectric absorption and Compton scattering are directly ionizing: kicking out an electron
- Photoelectric effect dominant at low energies, pair production at high energies



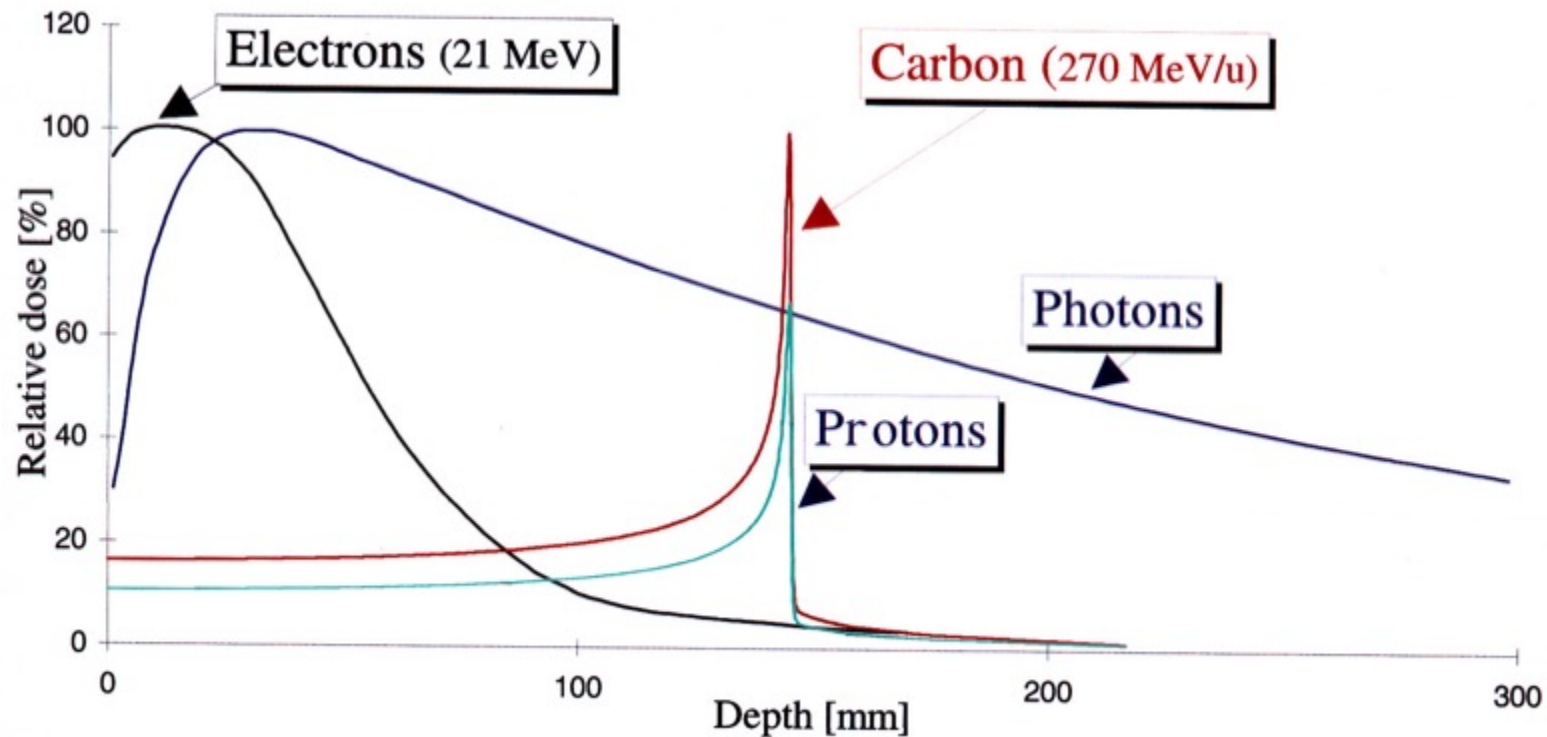
Charged particle interactions

- Coulomb scattering (elastic and inelastic)
 - Can consider scattering off of nucleus or atomic electrons (electron case is dominant)
 - Kinematics are different for heavy charged particle vs electron
 - Energy lost over many scatterings for heavier ions
- Production of Bremsstrahlung radiation
 - As particles decelerate, they emit radiation
 - Mostly a consideration for electrons, rapid deceleration
- Stopping power or linear energy transfer (LTE) quantifies energy lost per distance



Bragg peaks

- Very different energy deposition (and therefore dose) per depth for protons/heavy ions, electrons, and photons
- Localized peak makes protons/heavy ions desirable for radiation therapy



Nuclear interactions

- Generally write as some like projectile and heavy target, heavy and light products:
 - $a + X \rightarrow Y + b$; a and b are the light projectile and product
 - $X(a,b)Y$ form is a shorthand for classes of reactions, e.g. (n,p)
- Broad categories:
 - Scattering (a,a)
 - Knockout (a,Na)
 - Capture (a,γ)
 - Transfer (a,b)
 - Photonuclear reactions (γ,a)
 - Everything but scattering could be considered “transmutation”
- What reaction dominates is highly energy and isotope dependent

Nuclear scattering

- Nuclear scattering: scattering of nucleons/heavy ions with nucleus
 - For neutrons: strong nuclear force is *attractive* up to short distances, nucleus looks like a strong absorber
 - For protons/heavy ions: competes with Coulomb scattering
- Result looks like diffraction: analogy of the absorbing disc in optics
- Can be both elastic and inelastic
- Inelastic results in follow-on gamma emissions or decays

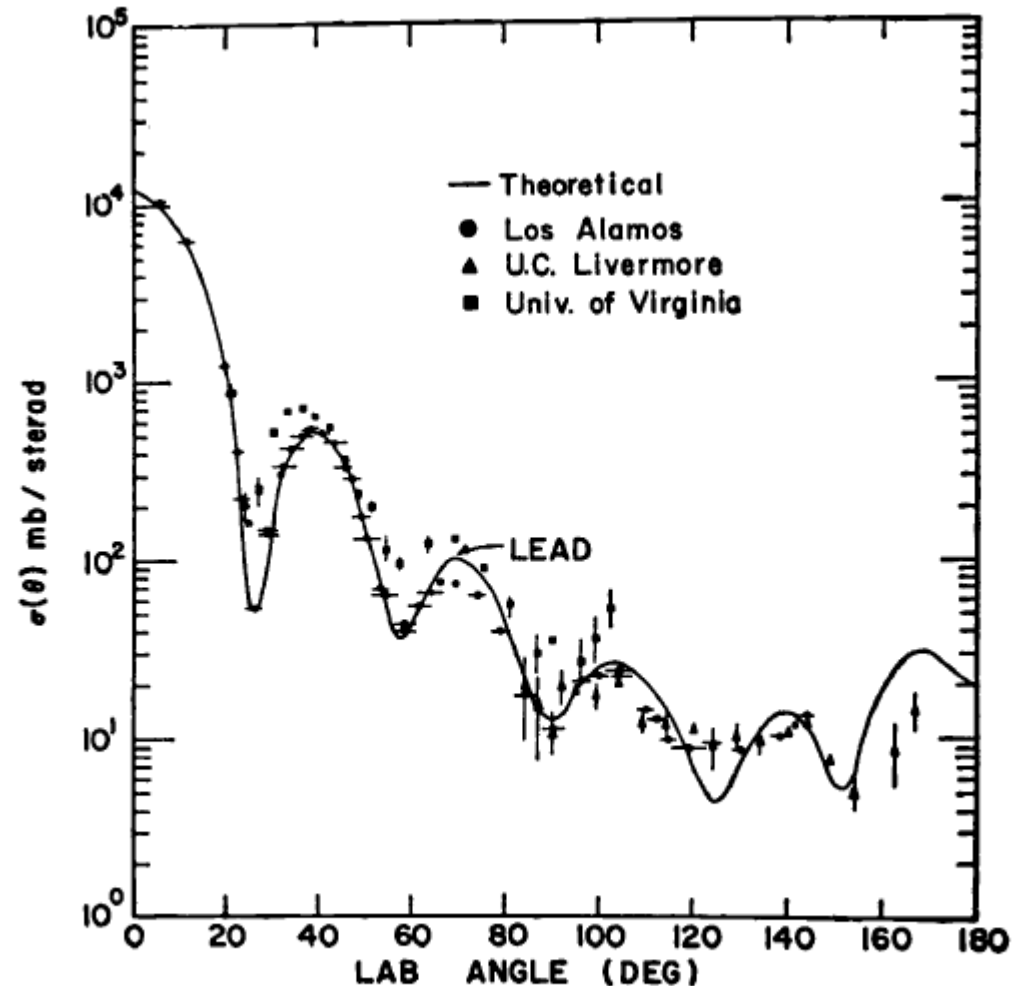


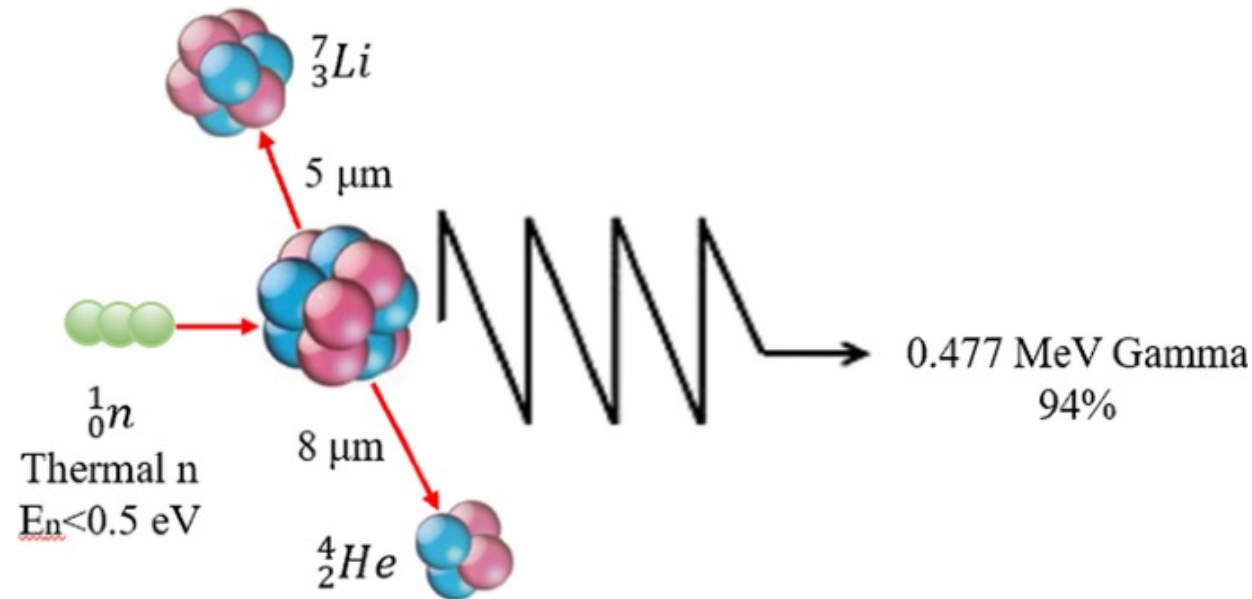
Figure 11.14 Elastic scattering of 14-MeV neutrons from Pb. From S. Fernbach, *Rev. Mod. Phys.* **30**, 414 (1958).

Knockout reactions

- Incoming nucleon (proton or neutron) collides with a nucleus, knocking out additional nucleons
- Generally a higher energy phenomena, e.g. ~ 10 MeV neutrons
- Common for neutron-rich nuclei, e.g. $(n,2n)$ and $(n,3n)$ reactions
- Also occurs for protons and ions with very high energies (e.g. colliders)
- Kinematics are complex: many-body problem
- Always the potential to leave nucleus in an excited state, resulting in gammas

Capture reactions

- Occurs when proton, neutron (or alpha particle) is absorbed into the nucleus, resulting in more energetically favorable nuclear state
 - In basic neutron capture, excess energy emitted as gamma
 - Can also include other products (fission)
- Especially important for lower-energy neutrons (eV-keV) down to “thermal” neutrons (0.025 eV)
- Can also create radioactive isotopes
- We utilize neutron capture to shield from/eliminate neutrons



Boron neutron capture therapy (BNCT) concept:

He *et al.*, *Radiat. Oncol.* (2021) **16**:216

<https://doi.org/10.1186/s13014-021-01939-7>

Transfer reactions

- Typically some sort of exchange of one nucleon for another, e.g. (p,n) or (n,p)
- Can also include alpha particles, e.g. (α ,n) or (n, α)
- Considered “direct” nuclear reactions:
 - Incident nucleon “sees” individual nucleons, i.e. its wavelength is on the scale of the nucleon (1 fm)
 - Contrast to compound nuclear reactions (temporary compound nucleus formation)

Reaction resonances

- Nuclear reaction cross-sections are highly energy dependent, especially at lower energies
- Every type of reaction can have resonances
- Design consideration: choose materials/dumps/etc. to avoid unwanted reaction resonances

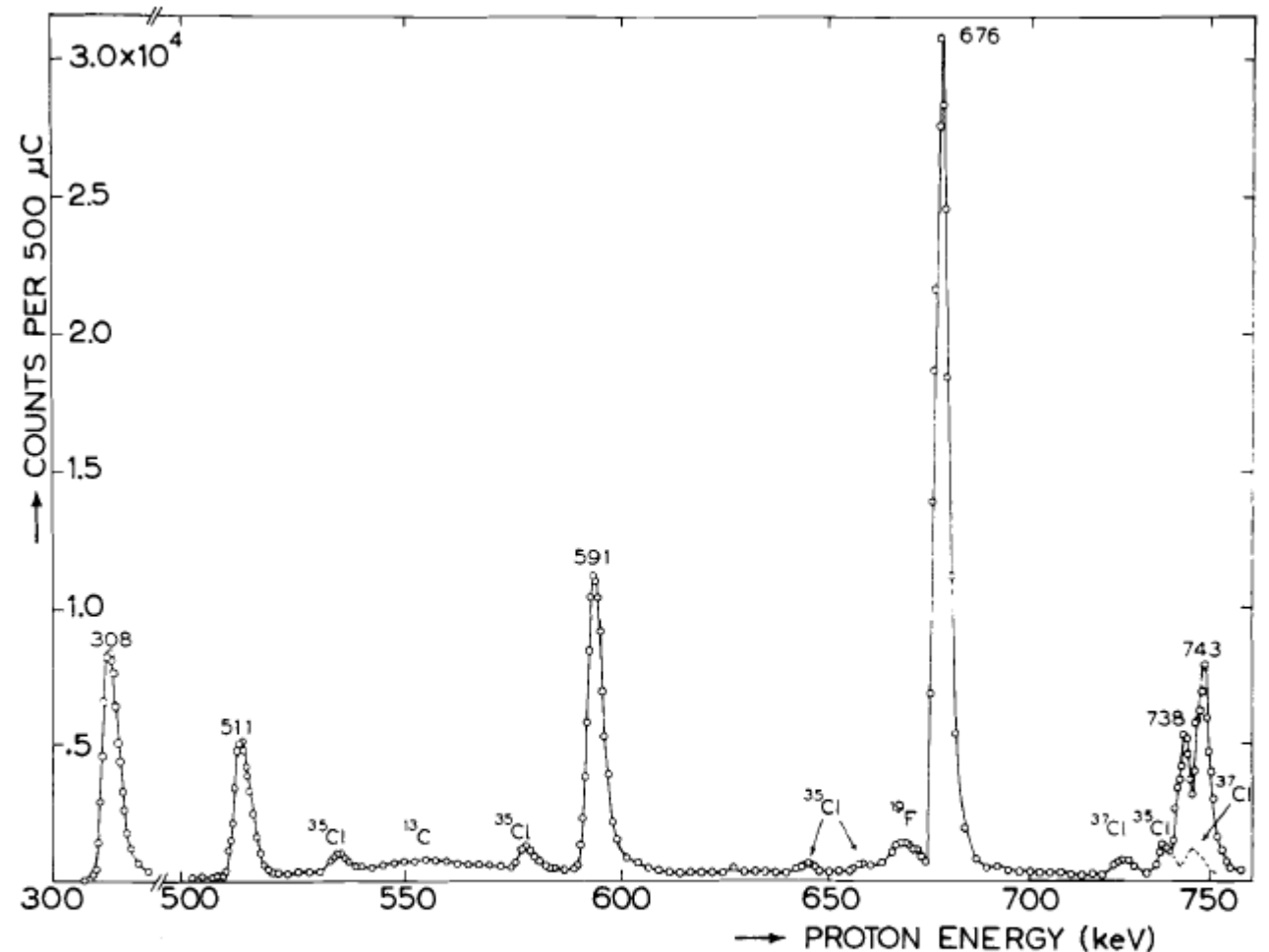


Figure 11.29 Resonances observed in the radiative proton capture by ^{23}Na . In this case, the total yield of γ rays in the energy range 3 – 13 MeV was measured as a function of the incident proton energy. The Cl peaks appear because the target used was NaCl. From P. W. M. Glaudemans and P. M. Endt, *Nucl. Phys.* **30**, 30 (1962).

Shielding

- Type of shielding depends on particle and energy
- Heavy ions/alpha particles:
 - Very easy to shield, short mean-free path due to higher mass
 - Tend to follow relatively straight particle “tracks” from kinematics
- Electrons/positrons:
 - Similarly, since they undergo Coulomb scattering, relatively easy to shield
 - Follow erratic paths, large scattering angles due to like masses
 - Secondary consideration of Bremsstrahlung radiation
- Gammas/X-rays:
 - Shielding largely requires high volume and/or density of material, especially for high energies
- Neutrons:
 - Much harder to shield, no electronic interactions
 - Biggest “mess” produced: activation, material damage

Radiation damage

Types of damage

- Ionization: production of ions by removing electrons
 - Happens both directly and indirectly from all kinds of (ionizing) radiation
 - Especially important in the biological case and for electronics
- Atomic displacement: knocking an atom out of its lattice
 - Mostly from neutron interactions, but other kinds of radiation can also cause displacements
 - Responsible for lots of damage phenomena, especially those affecting mechanical properties
 - Measure dose in “displacements per atom” dpa

Timescale of radiation damage

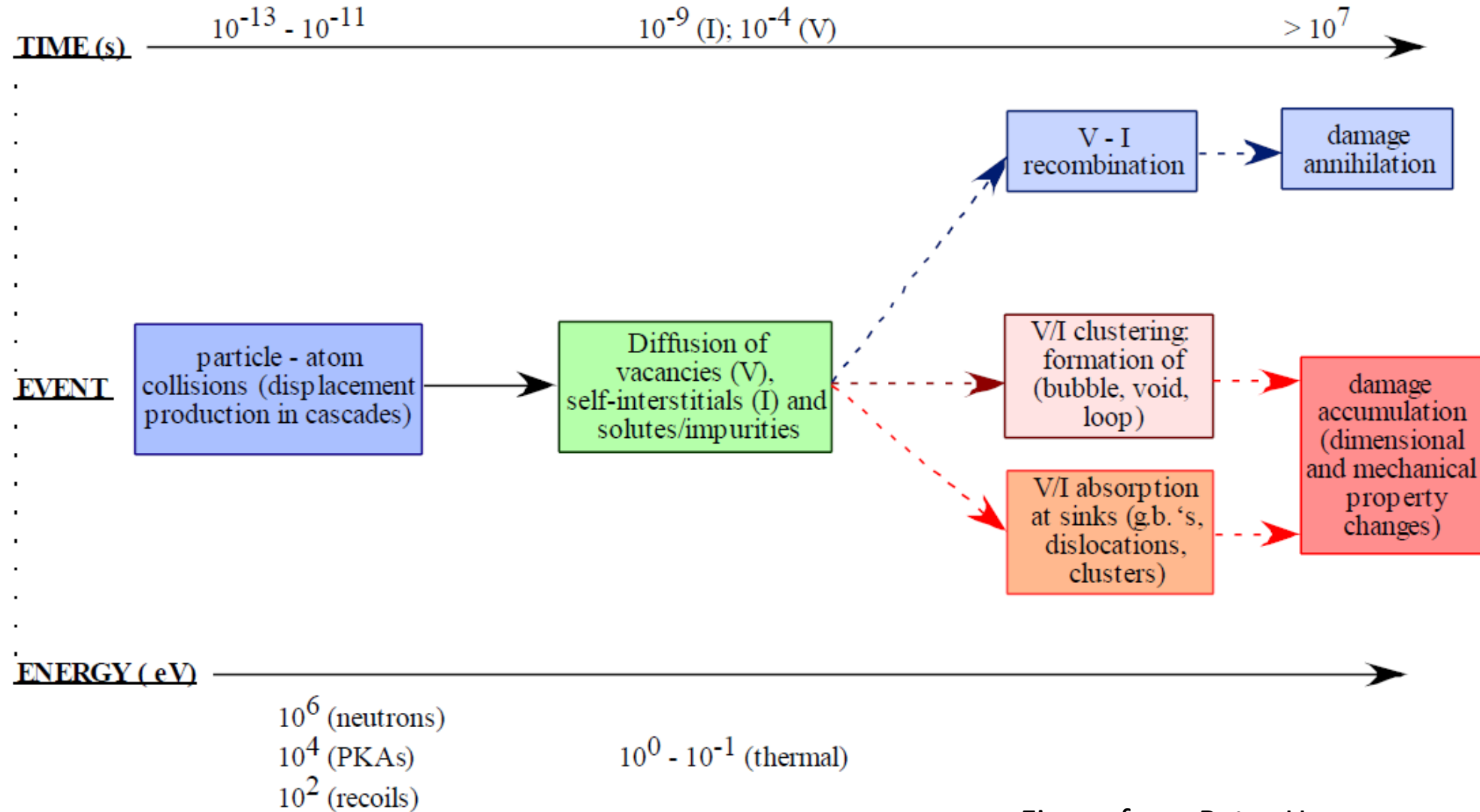


Figure from Peter Hosemann

Radiation damage in accelerators

- Under normal circumstances, biggest concerns are:
 - Immediate flux of radiation, i.e. X-rays while the accelerator is running
 - Activation of components, hardware, tools
 - Damage to objects beam directly interacts with, like beam dumps and targets
- Over time, must also consider:
 - Long term damage to structural components, magnets, electronics, detectors
- Additional possibility of beam loss
- Some previous USPAS classes:
 - Beam Loss and Machine Protection, January 2017
<https://uspas.fnal.gov/materials/17UCDavis/davis-machineprotection.shtml>
 - The Effect of Radiation on Electronics and Materials, June 2019
<https://uspas.fnal.gov/materials/19NewMexico/NewMexico-Radiation.shtml>

Effects of radiation

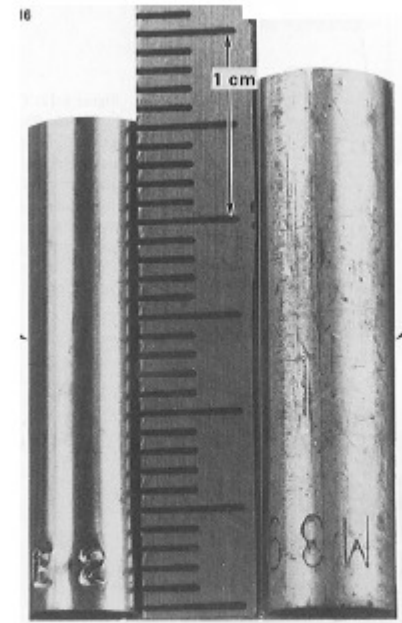
- Considerations for structural materials, magnets:
 - Swelling
 - Radiation hardening/embrittlement
 - Phase changes of materials (e.g. specific alloy phases)
 - Contamination from transmutation products, especially hydrogen and helium gas
 - Long-term and coupled effects like stress-corrosion cracking, creep



316L steel pipe after irradiation

D.L. Porter, F. A. Garner

J. Nuc. Mat. 159 (1988) 117-121



316L before and after irradiation

J.S. Straalsund, et al J. Nucl. Mater.

Figure from Peter Hosemann

Effects of radiation

- Radiation effects are very important for electronics:
 - Impact immediate performance (e.g. noise/sensitivity of detectors)
 - Damage materials: highly ordered crystal structures
- Broad area of study: radiation-hardening of electronics, e.g. for space applications

Radiation and superconductors

- In immediate case, radiation can ruin superconductivity
 - Deposition of heat, phonons
 - Creation of quasiparticles
 - Can lead to quench, which itself can cause damage
- In the long term, question of change in mechanical properties (e.g. embrittlement)
- Active area of study as new superconducting materials for magnets come into use

Using accelerators

- Accelerators are one of the primary ways we study radiation damage by creating it
 - High energy charged particle beams on targets (e.g. FRIB)
 - Production of neutron beams (e.g. SNS)
 - Study effects of certain amounts of dose on different materials
- Also use particle beams to view effects of radiation damage
 - Electron microscopes
 - Focused ion beams
- Accelerators are also used to produce radioisotopes for medical imaging and therapies